

DISCRETE MATHEMATICAL STRUCTURES

UNIT-1: MATHEMATICAL LOGIC

Dr. M. ANKAMMA RAO,

Assistant Professor,
Department of Mathematics & Statistics,
Vignan's Foundation for Science, Technology & Research (Deemed to be University)

July 21, 2026

Contents

1	Statements (Propositions)	4
1.1	Definition	4
1.2	Types of Statements	4
1.3	Examples of Statements	6
1.4	Applications of Statements	7
2	Logical Connectives	8
2.1	Types of Logical Connectives	8
2.2	Applications of Logical Connectives	11
3	Well-Formed Formulas (WFF)	11
3.1	Types of Well-Formed Formulas	12
3.2	Examples of Well-Formed Formulas	13
3.3	Examples	13
3.4	Applications of Well-Formed Formulas	15
4	Truth Tables	15
4.1	Types of Truth Tables	16
4.2	Applications of Truth Tables	19
5	Tautology	19
5.1	Types of Tautologies	20
5.2	Examples	21
5.3	Applications of Tautology	24
6	Logical Equivalence and Logical Implication	24
6.1	Logical Equivalence	24
6.2	Examples of Logical Equivalence	25
6.3	Applications of Logical Equivalence	29
6.4	Logical Implication	29
6.5	Types of Logical Implication	30
6.6	Examples of Logical Implication	31
6.7	Applications of Logical Implication	31

6.8	Difference Between Logical Equivalence and Logical Implication	32
7	Quantifiers	32
7.1	Types of Quantifiers	32
7.2	Universal Quantifier ($\forall$)	32
7.3	Examples	33
7.4	Existential Quantifier ($\exists$)	33
7.5	Examples	34
7.6	Unique Existential Quantifier ($\exists!$)	34
7.7	Examples	34
7.8	Applications of Quantifiers	35
8	Predicate Calculus (Predicate Logic)	36
8.1	Types of Predicate Calculus	38
8.2	First-Order Predicate Logic	38
8.3	2. Higher-Order Predicate Logic	38
8.4	Applications of Predicate Calculus	40
8.5	Difference Between Propositional Logic and Predicate Calculus	40
9	Inference Theory	40
9.1	Types of Inference	41
9.2	Rules of Inference in Propositional Logic	43
9.3	Applications of Inference Theory	45
10	Logic Gates and Their Operations	46
10.1	Types of Logic Gates	46
10.2	Basic Logic Gates	47
10.3	Universal and Special Logic Gates	48
10.4	Applications of Logic Gates	50
11	Normal Forms	50
11.1	Disjunctive Normal Form (DNF)	58
11.2	Principal Disjunctive Normal Form (PDNF)	64
11.3	Principal Conjunctive Normal Form (PCNF)	73
11.4	Conversion of CNF to DNF and vice versa	83
11.5	Applications of Normal Forms	88
12	Minimization of Boolean function (Karnaugh Maps)	88
12.1	Types of Karnaugh Maps	89
12.2	Steps for Karnaugh Map (K-Map) Simplification	89
12.3	Examples	92
12.4	Applications of Karnaugh Maps	109

1 Statements (Propositions)

1.1 Definition

A **statement** (also called a **proposition**) is a declarative sentence that has exactly one truth value. A statement is either **True (T)** or **False (F)**, but not both.

Statements are the fundamental building blocks of propositional logic.

Mathematically, a statement is represented by symbols such as

$$p, q, r, s, \dots$$

where each symbol denotes a proposition.

Example

Let

$$p : \text{“The Earth revolves around the Sun.”}$$

This statement is **True**.

Characteristics of a Statement

A valid statement must satisfy the following conditions.

- It must be a declarative sentence.
- It must have exactly one truth value.
- It must be either True or False.
- It should not be a question, command, or exclamation.

1.2 Types of Statements

Statements are commonly classified into the following categories.

1. Simple (Atomic) Statements
2. Compound Statements
3. Open Statements
4. Universal Statements
5. Existential Statements

1. Simple (Atomic) Statement

A **simple statement** consists of a single proposition and contains no logical connectives.

Examples

$$p : \text{“The classroom is open.”}$$

$$q : \text{“Today is Monday.”}$$

2. Compound Statement

A **compound statement** is formed by combining two or more simple statements using logical connectives such as AND, OR, NOT, IF–THEN, and IF AND ONLY IF.

Examples

$$p \wedge q$$

$$p \vee q$$

$$p \rightarrow q$$

$$p \leftrightarrow q$$

3. Open Statement

An **open statement** contains one or more variables whose truth value depends on the values assigned to those variables.

Example

$$x + 5 = 12$$

This is not a statement until a value is assigned to x .

If

$$x = 7,$$

then the statement becomes True.

4. Universal Statement

A universal statement asserts that a property holds for every element of a set.

It uses the universal quantifier

$$\forall$$

which means “for all.”

Example

$$\forall x \in \mathbb{R}, x^2 \geq 0.$$

5. Existential Statement

An existential statement asserts that there exists at least one element satisfying a given property.

It uses the existential quantifier

$$\exists$$

which means “there exists.”

Example

$$\exists x \in \mathbb{R} \text{ such that } x^2 = 25.$$

1.3 Examples of Statements

Table 1.1: Examples of Valid Statements

Statement	Truth Value
The Sun rises in the East.	True
India is in Asia.	True
The Earth has two moons.	False
5 is an even number.	False
A triangle has three sides.	True

Examples of Non-Statements

The following expressions are **not** statements because they do not possess a definite truth value.

Table 1.2: Examples of Non-Statements

Expression	Reason
What is your name?	Question
Please close the door.	Command
Study regularly.	Instruction
Wow! What a beautiful day!	Exclamation
$x + 3 = 7$	Depends on the value of x

Examples

Example 1: Weather

Let

$$p : \text{“It is raining.”}$$

This statement is

- True, if it is raining.
- False, otherwise.

Example 2: Banking

Let

$$q : \text{“The account balance exceeds ₹10,000.”}$$

If the balance is ₹15,000,

$$q = \text{True.}$$

Otherwise,

$$q = \text{False.}$$

Example 3: Healthcare

Let

r : “The patient’s body temperature is above 38°C.”

The statement is either True or False depending on the measured temperature.

Example 4: Education

Let

s : “The student passed the examination.”

The truth value depends on the examination result.

Example 5: Online Shopping

Let

t : “The ordered product has been delivered.”

The statement becomes True after successful delivery.

1.4 Applications of Statements

Statements play an important role in various fields of science and engineering.

1. Mathematics

Statements are used to formulate definitions, theorems, proofs, and logical arguments.

2. Computer Programming

Conditional statements such as `if`, `else`, and `while` are based on logical propositions.

3. Artificial Intelligence

Knowledge representation and logical reasoning rely on statements.

4. Digital Electronics

Logic gates process binary statements represented by True and False.

5. Database Systems

SQL queries evaluate logical statements using operators such as AND, OR, and NOT.

6. Cyber Security

Authentication systems verify statements such as “Username is valid” and “Password is correct.”

7. Decision Support Systems

Business rules are expressed as logical statements for automated decision making.

8. Software Testing

Logical statements are evaluated to verify program correctness.

2 Logical Connectives

Definition

Logical connectives are words or symbols used to combine one or more logical statements to form a new compound statement. The truth value of the compound statement depends on the truth values of the individual statements.

For example, let

p : “It is raining.”

q : “I carry an umbrella.”

Using logical connectives, we can form compound statements such as:

$$p \wedge q, \quad p \vee q, \quad \neg p, \quad p \rightarrow q, \quad p \leftrightarrow q.$$

2.1 Types of Logical Connectives

There are five basic logical connectives.

1. Negation ($\neg$)
2. Conjunction ($\wedge$)
3. Disjunction ($\vee$)
4. Implication (Conditional) ($\rightarrow$)
5. Biconditional ($\leftrightarrow$)

1. Negation ($\neg$)

Definition:

Negation reverses the truth value of a statement.

If

p : “It is raining.”

then

$\neg p$: “It is not raining.”

Symbol:

$$\neg p$$

Example

Let

p : “The classroom projector is ON.”

Then

$\neg p$: “The classroom projector is OFF.”

2. Conjunction ($\wedge$)

Definition:

The conjunction of two statements is true only when both statements are true.

Symbol:

$$p \wedge q$$

which is read as

$$“p \text{ AND } q”$$

Example

Let

$$p : \text{“The ATM card is valid.”}$$

$$q : \text{“The correct PIN is entered.”}$$

Then

$$p \wedge q :$$

“The ATM card is valid AND the correct PIN is entered.”

The transaction succeeds only if both conditions are true.

3. Disjunction ($\vee$)

Definition:

The disjunction of two statements is true if at least one of the statements is true.

Symbol:

$$p \vee q$$

which is read as

$$“p \text{ OR } q”$$

Example

Let

$$p : \text{“The meeting is conducted offline.”}$$

$$q : \text{“The meeting is conducted online.”}$$

Then

$$p \vee q$$

means

“The meeting is conducted offline OR online.”

The statement is true if at least one mode is available.

4. Implication (Conditional) ($\rightarrow$)

Definition:

An implication states that if the first statement is true, then the second statement must also be true.

Symbol:

$$p \rightarrow q$$

which is read as

“If p , then q .”

Example

Let

p : “A student scores more than 40 marks.”

q : “The student passes the examination.”

Then

$$p \rightarrow q$$

means

“If a student scores more than 40 marks, then the student passes the examination.”

5. Biconditional ($\leftrightarrow$)

Definition:

A biconditional statement is true when both statements have the same truth value (both true or both false).

Symbol:

$$p \leftrightarrow q$$

which is read as

“ p if and only if q .”

Example

Let

p : “You can enter the laboratory.”

q : “You possess a valid ID card.”

Then

$$p \leftrightarrow q$$

means

“You can enter the laboratory if and only if you possess a valid ID card.”

Summary of Logical Connectives

Table 2.1: Basic Logical Connectives

Connective	Symbol	Meaning
Negation	$\neg$	NOT
Conjunction	$\wedge$	AND
Disjunction	$\vee$	OR
Implication	$\rightarrow$	IF ... THEN
Biconditional	$\leftrightarrow$	IF AND ONLY IF

2.2 Applications of Logical Connectives

Logical connectives are widely used in:

- Digital circuit design
- Computer programming
- Artificial Intelligence
- Database query processing
- Decision-making systems
- Mathematical proofs
- Search engines and recommendation systems

3 Well-Formed Formulas (WFF)

Definition

A **Well-Formed Formula (WFF)** is a logical expression that is constructed according to the syntactic rules of propositional logic. Every WFF has a unique logical meaning and can be assigned a truth value (True or False).

In simple terms, a WFF is a **grammatically correct logical expression** formed using propositional variables, logical connectives, and parentheses.

For example,

$$(p \wedge q), \quad (p \rightarrow q), \quad \neg(p \vee q)$$

are well-formed formulas because they follow the rules of propositional logic.

On the other hand,

$$\wedge pq, \quad p \rightarrow, \quad (p \wedge)$$

are **not** well-formed formulas because they violate the syntax rules.

Formation Rules of Well-Formed Formulas

A logical expression is a Well-Formed Formula if it satisfies the following rules.

1. Every propositional variable such as

$$p, q, r, s$$

is a WFF.

2. If A is a WFF, then

$$\neg A$$

is also a WFF.

3. If A and B are WFFs, then each of the following is also a WFF:

$$(A \wedge B)$$

$$(A \vee B)$$

$$(A \rightarrow B)$$

$$(A \leftrightarrow B)$$

4. Parentheses must be placed correctly to remove ambiguity.

3.1 Types of Well-Formed Formulas

Although WFFs are not formally classified into fixed categories, they are commonly grouped according to their structure.

1. Atomic Well-Formed Formula

An atomic WFF consists of only one propositional variable.

$$p, \quad q, \quad r$$

Example

Let

$$p : \text{“The classroom is open.”}$$

Since it contains only one statement,

$$p$$

is an atomic WFF.

2. Compound Well-Formed Formula

A compound WFF is formed by combining two or more atomic statements using logical connectives.

Example:

$$(p \wedge q)$$

$$(p \rightarrow q)$$

$$\neg(p \vee q)$$

Example

Let

p : “The internet is available.”

q : “Students can attend the online class.”

Then

$$p \rightarrow q$$

means

“If the internet is available, then students can attend the online class.”

3.2 Examples of Well-Formed Formulas

Table 3.1: Examples of WFFs

Formula	Status
p	WFF
$\neg p$	WFF
$(p \wedge q)$	WFF
$(p \vee q)$	WFF
$(p \rightarrow q)$	WFF
$(p \leftrightarrow q)$	WFF
$\neg(p \wedge q)$	WFF
$((p \rightarrow q) \wedge r)$	WFF

Examples of Expressions that are NOT WFFs

Table 3.2: Examples of Invalid Logical Expressions

Expression	Reason
$p \wedge$	Missing second statement
$\rightarrow q$	Missing first statement
$\wedge pq$	Incorrect syntax
$(p \vee q$	Missing closing parenthesis
pq	No logical connective

3.3 Examples

Example 1: Banking

Let

p : “The ATM card is valid.”

q : “The correct PIN is entered.”

Then

$$(p \wedge q)$$

means

“The ATM card is valid AND the correct PIN is entered.”

The transaction is successful only if both conditions are satisfied.

Example 2: Healthcare

Let

p : “The patient’s temperature is above 38°C.”

q : “The doctor prescribes medication.”

Then

$$p \rightarrow q$$

means

“If the patient’s temperature is above 38°C, then the doctor prescribes medication.”

Example 3: University

Let

p : “The student has paid the examination fee.”

q : “The student receives the hall ticket.”

Then

$$p \rightarrow q$$

represents the examination rule.

Example 4: Smart Home

Let

p : “Motion is detected.”

q : “The security alarm is activated.”

Then

$$p \rightarrow q$$

represents the automation rule.

3.4 Applications of Well-Formed Formulas

Well-Formed Formulas are widely used in many fields.

1. Computer Programming

Logical conditions in programming languages such as C, C++, Java, and Python are represented using WFFs.

Example:

$$(\text{Age} \geq 18) \wedge (\text{Citizen} = \text{True})$$

determines whether a person is eligible to vote.

2. Artificial Intelligence

AI systems use WFFs to represent knowledge and make logical decisions.

3. Digital Electronics

Logic gates such as AND, OR, and NOT are designed using well-formed logical expressions.

4. Database Systems

SQL queries use logical expressions to retrieve desired records.

Example:

$$(\text{Salary} > 50000) \wedge (\text{Department} = \text{'CSE'})$$

5. Expert Systems

Medical diagnosis systems and decision-support systems use WFFs to represent rules.

6. Cyber Security

Access control systems verify multiple conditions before granting user access.

7. Mathematical Proofs

Mathematicians use WFFs to construct valid logical arguments and proofs.

4 Truth Tables

Definition

A **Truth Table** is a tabular representation that lists all possible truth values of one or more logical statements and shows the resulting truth value of a compound logical expression.

Truth tables are used to analyze logical expressions, verify logical equivalence, determine the validity of arguments, and design digital circuits.

For example, let

$$p : \text{"It is raining."}$$

$$q : \text{"I carry an umbrella."}$$

The truth table of the compound statement

$$p \wedge q$$

shows the truth value for every possible combination of p and q .

4.1 Types of Truth Tables

Truth tables are commonly classified according to the number of propositions involved.

- 1. Truth Table for a Single Proposition
- 2. Truth Table for Two Propositions
- 3. Truth Table for Three or More Propositions
- 4. Truth Tables for Logical Connectives

1. Truth Table for a Single Proposition

A single proposition has only two possible truth values.

p

Table 4.1: Truth Table for a Single Proposition

p
T
F

Example

Let

p : “The classroom light is ON.”

If the light is ON, then

$p = T$

otherwise,

$p = F$.

2. Truth Table for Two Propositions

For two propositions, there are

$2^2 = 4$

possible combinations.

Let

p : “The ATM card is valid.”

q : “The correct PIN is entered.”

Table 4.2: Truth Values of Two Propositions

p	q
T	T
T	F
F	T
F	F

3. Truth Table for Three Propositions

Three propositions produce

$$2^3 = 8$$

possible combinations.
Let

$$p, q, r$$

be three propositions.

Table 4.3: Truth Values of Three Propositions

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Truth Tables of Basic Logical Connectives

1. Negation ($\neg$)

$$\neg p$$

Table 4.4: Truth Table of Negation

p	$\neg p$
T	F
F	T

Example

Let

$$p : \text{“The internet is available.”}$$

Then

$$\neg p : \text{“The internet is not available.”}$$

2. Conjunction ($\wedge$)

$$p \wedge q$$

Table 4.5: Truth Table of Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example

A student receives a scholarship only if

- the student passes the examination, and
- the student's attendance is at least 75%.

3. Disjunction ($\vee$)

$$p \vee q$$

Table 4.6: Truth Table of Disjunction

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example

A meeting can be attended

- offline, or
- online.

If at least one option is available, participation is possible.

4. Implication ($\rightarrow$)

$$p \rightarrow q$$

Table 4.7: Truth Table of Implication

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Example

If a student pays the examination fee, then the student receives the hall ticket.

5. Biconditional ($\leftrightarrow$)

$$p \leftrightarrow q$$

Table 4.8: Truth Table of Biconditional

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example

A laboratory door opens if and only if the user's ID card is valid.

4.2 Applications of Truth Tables

Truth tables are widely used in many areas of science and engineering.

1. Digital Electronics

Truth tables describe the operation of logic gates such as AND, OR, NOT, NAND, NOR, XOR, and XNOR.

2. Computer Programming

Logical conditions in programming languages are tested using truth tables.

3. Artificial Intelligence

Expert systems use truth tables to make logical decisions.

4. Database Systems

Logical operators such as AND, OR, and NOT are evaluated using truth tables in SQL queries.

5. Mathematical Logic

Truth tables verify logical equivalence, tautologies, contradictions, and the validity of arguments.

6. Control Systems

Automatic systems such as traffic signals, elevators, and industrial controllers use truth tables to define decision rules.

7. Cyber Security

Authentication systems verify multiple conditions before granting access.

8. Communication Systems

Truth tables are used in error detection and digital communication protocols.

5 Tautology

Definition

A **Tautology** is a compound logical statement that is **always true**, regardless of the truth values of its constituent propositions. In other words, a tautology evaluates to **True (T)** for every possible combination of truth values of the individual statements. Mathematically,

If A is a logical formula, then A is a tautology if $A \equiv T$.

Example

The statement

$$p \vee \neg p$$

is a tautology because either p is true or its negation $\neg p$ is true.

Truth Table Verification

Consider

$$p \vee \neg p$$

Table 5.1: Truth Table of a Tautology

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Since the final column contains only **True**, the expression is a tautology.

5.1 Types of Tautologies

Although there is no universally accepted formal classification, tautologies are commonly grouped into the following categories for teaching and practical applications.

1. Basic Tautologies
2. Logical Law Tautologies
3. Conditional Tautologies
4. Biconditional Tautologies

1. Basic Tautologies

These are the simplest tautologies.

Examples:

$$p \vee \neg p$$

$$\neg(p \wedge \neg p)$$

2. Logical Law Tautologies

These arise from fundamental laws of propositional logic.

Examples:

$$(p \rightarrow q) \equiv (\neg p \vee q)$$

$$(p \wedge q) \rightarrow p$$

$$(p \wedge q) \rightarrow q$$

3. Conditional Tautologies

These involve implication.

Examples:

$$p \rightarrow p$$

$$(p \wedge q) \rightarrow p$$

$$(p \wedge q) \rightarrow q$$

4. Biconditional Tautologies

These involve logical equivalence.

Example:

$$(p \leftrightarrow q) \leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow p)]$$

5.2 Examples

Example 1: University Attendance

Let

$$p : \text{“A student is present.”}$$

Then

$$p \vee \neg p$$

means

“A student is either present or absent.”

This statement is always true because every student must be either present or absent.

Example 2: Weather

Let

$$p : \text{“It is raining.”}$$

Then

$$p \vee \neg p$$

means

“Either it is raining or it is not raining.”

This is always true.

Example 3:

Determine whether the following compound proposition is a tautology:

$$(P \rightarrow Q) \vee (Q \rightarrow P).$$

Solution

Step 1: Write the Given Proposition

The given proposition is

$$F = (P \rightarrow Q) \vee (Q \rightarrow P).$$

Step 2: Construct the Truth Table

Recall that

$$P \rightarrow Q \equiv \bar{P} \vee Q.$$

Construct the truth table.

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \vee (Q \rightarrow P)$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	1	1	1

Step 3: Analyze the Final Column

The final column is

$$1, 1, 1, 1.$$

Since the proposition is true for every possible combination of truth values of P and Q ,

$$(P \rightarrow Q) \vee (Q \rightarrow P)$$

is always true.

Step 4: Conclusion

A compound proposition that is true for every possible assignment of truth values is called a **tautology**.

Hence,

$$(P \rightarrow Q) \vee (Q \rightarrow P) \text{ is a tautology.}$$

Final Answer

$$(P \rightarrow Q) \vee (Q \rightarrow P) \text{ is a tautology.}$$

Example 4:

Problem:

Using a truth table, determine whether the following compound proposition is a tautology:

$$(P \rightarrow Q) \leftrightarrow (\bar{Q} \rightarrow \bar{P}).$$

Solution

Step 1: Write the Given Proposition

The given proposition is

$$F = (P \rightarrow Q) \leftrightarrow (\bar{Q} \rightarrow \bar{P}).$$

Step 2: Construct the Truth Table

Recall that

$$P \rightarrow Q \equiv \bar{P} \vee Q.$$

Also,

$$\bar{Q} \rightarrow \bar{P} \equiv Q \vee \bar{P}.$$

Construct the truth table.

P	Q	$\bar{P}$	$\bar{Q}$	$P \rightarrow Q$	$\bar{Q} \rightarrow \bar{P}$	F
0	0	1	1	1	1	1
0	1	1	0	1	1	1
1	0	0	1	0	0	1
1	1	0	0	1	1	1

Step 3: Analyze the Final Column

The final column is

$$1, 1, 1, 1.$$

Since every entry in the final column is 1, the proposition is true for every possible assignment of truth values.

Step 4: Conclusion

Therefore,

$$(P \rightarrow Q) \leftrightarrow (\bar{Q} \rightarrow \bar{P})$$

is a tautology.

This result is known as the **Law of Contraposition**, which states that a statement and its contrapositive are logically equivalent.

Final Answer

$$(P \rightarrow Q) \leftrightarrow (\bar{Q} \rightarrow \bar{P}) \text{ is a tautology.}$$

5.3 Applications of Tautology

Tautologies have numerous applications in mathematics, computer science, and engineering.

1. **Mathematical Proofs**

Tautologies are used to prove logical identities and mathematical theorems.

2. **Digital Electronics**

Boolean expressions are simplified using tautological laws while designing digital circuits.

3. **Computer Programming**

Compilers and program verification tools use tautologies to optimize logical conditions.

4. **Artificial Intelligence**

Knowledge-based systems use tautologies to ensure that logical inference rules remain valid.

5. **Database Systems**

Query optimizers simplify SQL conditions using tautological expressions.

6. **Cyber Security**

Authentication systems verify logical access-control rules using tautological identities.

7. **Formal Verification**

Hardware and software verification tools use tautologies to prove system correctness.

8. **Decision Support Systems**

Business rule engines verify that decision rules are logically consistent.

6 Logical Equivalence and Logical Implication

Logical Equivalence and Logical Implication are two important concepts in propositional logic. They are widely used in mathematics, computer science, artificial intelligence, digital electronics, and automated reasoning.

6.1 Logical Equivalence

Definition

Two logical statements are said to be **logically equivalent** if they have the **same truth value for every possible combination** of truth values of their constituent propositions.

If two statements P and Q are logically equivalent, we write

$$P \equiv Q.$$

This means that the truth tables of P and Q are identical.

Truth Table Verification

Consider

$$P \rightarrow Q$$

and

$$\neg P \vee Q.$$

Table 6.1: Verification of Logical Equivalence

P	Q	$P \rightarrow Q$	$\neg P \vee Q$	Equivalent
T	T	T	T	Yes
T	F	F	F	Yes
F	T	T	T	Yes
F	F	T	T	Yes

Since the last two columns are identical,

$$P \rightarrow Q \equiv \neg P \vee Q.$$

Common Logical Equivalences

Table 6.2: Common Logical Equivalence Laws

Law	Logical Equivalence
Identity Law	$P \wedge T \equiv P, \quad P \vee F \equiv P$
Domination Law	$P \vee T \equiv T, \quad P \wedge F \equiv F$
Idempotent Law	$P \vee P \equiv P, \quad P \wedge P \equiv P$
Double Negation	$\neg(\neg P) \equiv P$
Commutative Law	$P \vee Q \equiv Q \vee P$ $P \wedge Q \equiv Q \wedge P$
Associative Law	$(P \vee Q) \vee R \equiv P \vee (Q \vee R)$ $(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$
Distributive Law	$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$ $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$
De Morgan's Laws	$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$
Implication Law	$P \rightarrow Q \equiv \neg P \vee Q$
Biconditional Law	$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$

6.2 Examples of Logical Equivalence

Example 1: Banking

Let

P : “The ATM card is valid.”

Q : “The transaction is permitted.”

The statement

$$P \rightarrow Q$$

is logically equivalent to

$$\neg P \vee Q.$$

Example 2: Online Learning

Let

$$P : \text{“Internet is available.”}$$

$$Q : \text{“Students can attend the online class.”}$$

The implication

$$P \rightarrow Q$$

can also be written as

$$\neg P \vee Q.$$

Example 3:

Show that the following two propositions are logically equivalent:

$$P \rightarrow (Q \wedge R)$$

and

$$(P \rightarrow Q) \wedge (P \rightarrow R).$$

Verify the result

1. using logical equivalence laws, and
2. using a truth table.

Solution

Method 1: Using Logical Equivalence Laws

Consider

$$P \rightarrow (Q \wedge R).$$

Using the Implication Law,

$$P \rightarrow (Q \wedge R) \equiv \bar{P} \vee (Q \wedge R).$$

Applying the Distributive Law,

$$\bar{P} \vee (Q \wedge R) \equiv (\bar{P} \vee Q) \wedge (\bar{P} \vee R).$$

Again, using the Implication Law,

$$\bar{P} \vee Q \equiv P \rightarrow Q,$$

and

$$\bar{P} \vee R \equiv P \rightarrow R.$$

Hence,

$$P \rightarrow (Q \wedge R) \equiv (P \rightarrow Q) \wedge (P \rightarrow R).$$

Therefore, the two propositions are logically equivalent.

Method 2: Using the Truth Table

Construct the truth table.

P	Q	R	$Q \wedge R$	$P \rightarrow (Q \wedge R)$	$(P \rightarrow Q) \wedge (P \rightarrow R)$
0	0	0	0	1	1
0	0	1	0	1	1
0	1	0	0	1	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	0	0	0
1	1	1	1	1	1

Comparison

Observe that the last two columns are identical.

$P \rightarrow (Q \wedge R)$	$(P \rightarrow Q) \wedge (P \rightarrow R)$
1	1
1	1
1	1
1	1
0	0
0	0
0	0
1	1

Since the corresponding truth values are identical for every possible assignment of P , Q , and R ,

$$P \rightarrow (Q \wedge R) \equiv (P \rightarrow Q) \wedge (P \rightarrow R).$$

Conclusion

Both methods give the same result.

Hence,

$$P \rightarrow (Q \wedge R) \equiv (P \rightarrow Q) \wedge (P \rightarrow R).$$

Therefore, the given propositions are logically equivalent.

Final Answer

$$P \rightarrow (Q \wedge R) \equiv (P \rightarrow Q) \wedge (P \rightarrow R).$$

Example 4:

Problem:

Show that the following two propositions are logically equivalent:

$$\neg(P \vee Q)$$

and

$$\neg P \wedge \neg Q.$$

Verify the result

1. using logical equivalence laws, and
2. using a truth table.

Solution

Method 1: Using Logical Equivalence Laws

Consider

$$\neg(P \vee Q).$$

Using **De Morgan's Law**,

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q.$$

Hence,

$$\boxed{\neg(P \vee Q) \equiv \neg P \wedge \neg Q.}$$

Therefore, the two propositions are logically equivalent.

Method 2: Using the Truth Table

Construct the truth table.

P	Q	$P \vee Q$	$\neg(P \vee Q)$	$\neg P$	$\neg P \wedge \neg Q$
0	0	0	1	1	1
0	1	1	0	1	0
1	0	1	0	0	0
1	1	1	0	0	0

Comparison

Compare the last two columns.

$\neg(P \vee Q)$	$\neg P \wedge \neg Q$
1	1
0	0
0	0
0	0

The truth values are identical for every possible assignment of P and Q .

Hence,

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q.$$

Conclusion

Since both propositions have identical truth values in every row of the truth table and satisfy De Morgan's Law,

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q.$$

Therefore, the given propositions are logically equivalent.

Final Answer

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q.$$

6.3 Applications of Logical Equivalence

1. Simplifying logical expressions.
2. Designing digital circuits.
3. Optimizing computer programs.
4. Database query optimization.
5. Automated theorem proving.
6. Artificial intelligence reasoning.
7. Formal software verification.

6.4 Logical Implication

Definition

A statement P **logically implies** another statement Q if whenever P is true, Q must also be true.

Logical implication is written as

$$P \models Q.$$

It means

Whenever P is true, Q must also be true.

Equivalently,

$$P \models Q \iff P \rightarrow Q \text{ is a tautology.}$$

Truth Table Verification

Suppose

$$P = (p \wedge q)$$

and

$$Q = p.$$

Table 6.3: Verification of Logical Implication

p	q	$P = (p \wedge q)$	$Q = p$	$P \rightarrow Q$
T	T	T	T	T
T	F	F	T	T
F	T	F	F	T
F	F	F	F	T

Since

$$P \rightarrow Q$$

is always true,

$$(p \wedge q) \models p.$$

6.5 Types of Logical Implication

Logical implication is commonly categorized as follows.

1. Direct Implication
2. Contrapositive Implication
3. Hypothetical Implication
4. Mathematical (Logical) Implication

1. Direct Implication

$$P \rightarrow Q$$

Example:

”If a student scores above 40 marks, then the student passes.”

2. Contrapositive Implication

The contrapositive of

$$P \rightarrow Q$$

is

$$\neg Q \rightarrow \neg P.$$

Both statements are logically equivalent.

Example:

”If a student does not pass, then the student did not score above 40 marks.”

3. Hypothetical Implication

If

$$P \rightarrow Q$$

and

$$Q \rightarrow R,$$

then

$$P \rightarrow R.$$

Example:

If a student studies, then the student passes.

If the student passes, then the student graduates.

Therefore,

If the student studies, then the student graduates.

4. Mathematical (Logical) Implication

Example:

$$x > 5 \models x > 2.$$

6.6 Examples of Logical Implication

Example 1: Banking

If the correct PIN is entered,
then the ATM transaction is authorized.

Example 2: Healthcare

If a patient's temperature exceeds 38°C,
then the doctor recommends medication.

Example 3: Smart Home

If motion is detected,
then the security alarm is activated.

Example 4: University

If the examination fee is paid,
then the hall ticket is issued.

6.7 Applications of Logical Implication

1. Mathematical theorem proving.
2. Artificial intelligence inference engines.
3. Rule-based expert systems.
4. Compiler optimization.
5. Database integrity constraints.
6. Decision-support systems.

7. Cyber security access-control rules.
8. Automated reasoning and formal verification.

6.8 Difference Between Logical Equivalence and Logical Implication

Table 6.4: Logical Equivalence vs. Logical Implication

Feature	Logical Equivalence	Logical Implication
Meaning	Two statements always have identical truth values.	One statement guarantees the truth of another.
Notation	$P \equiv Q$	$P \models Q$
Truth Table	Final columns are identical.	$P \rightarrow Q$ is a tautology.
Relationship	Bidirectional.	Unidirectional.
Example	$P \rightarrow Q \equiv \neg P \vee Q$	$(P \wedge Q) \models P$

7 Quantifiers

Definition

A **quantifier** is a logical operator used in predicate logic to specify the quantity of objects in a given domain for which a predicate is true. Quantifiers indicate whether a statement applies to **all** elements or to **at least one** element of a set.

Suppose $P(x)$ is a predicate defined on a domain D . Then quantifiers allow us to express statements such as

$$\forall x \in D, P(x)$$

or

$$\exists x \in D, P(x).$$

Quantifiers extend propositional logic to predicate logic by introducing variables and specifying the scope of predicates.

7.1 Types of Quantifiers

There are two fundamental quantifiers in predicate logic.

1. Universal Quantifier
2. Existential Quantifier

Some textbooks also discuss the **Unique Existential Quantifier** as an extension.

7.2 Universal Quantifier ($\forall$)

Definition

The **Universal Quantifier** states that a predicate is true for **every** element in the domain.

It is denoted by

$$\forall$$

and is read as

“For all” or “For every.”

If $P(x)$ is a predicate, then

$$\forall x \in D, P(x)$$

means that $P(x)$ is true for every element of the domain D .

Mathematical Example

$$\forall x \in \mathbb{R}, x^2 \geq 0.$$

This statement is true because the square of every real number is non-negative.

7.3 Examples

Example 1: Education

$\forall x$, “Every registered student has a roll number.”

Example 2: Traffic Rules

$\forall x$, “Every vehicle must obey traffic signals.”

Example 3: Banking

$\forall x$, “Every customer must maintain a valid account.”

7.4 Existential Quantifier ($\exists$)

Definition

The **Existential Quantifier** states that there exists **at least one** element in the domain for which the predicate is true.

It is denoted by

$$\exists$$

and is read as

“There exists.”

If $P(x)$ is a predicate, then

$$\exists x \in D, P(x)$$

means that there is at least one element in D satisfying $P(x)$.

Mathematical Example

$$\exists x \in \mathbb{R} \text{ such that } x^2 = 25.$$

This statement is true because

$$x = 5$$

or

$$x = -5.$$

7.5 Examples

Example 1: University

”There exists a student who scored 100 marks.”

$\exists x, P(x)$

Example 2: Hospital

”There exists a patient requiring emergency treatment.”

Example 3: Banking

”There exists a customer eligible for a loan.”

7.6 Unique Existential Quantifier ($\exists!$)

Definition

The **Unique Existential Quantifier** states that there exists **exactly one** element satisfying the given predicate. It is denoted by

$\exists!$

and is read as

“There exists exactly one.”

Mathematical Example

$\exists! x \in \mathbb{N}$ such that $x + 5 = 8$.

Only

$x = 3$

satisfies the equation.

7.7 Examples

Example 1: College

”There exists exactly one Principal of the college.”

Example 2: Company

”There exists exactly one Chief Executive Officer (CEO).”

Example 3: Cricket Team

”There exists exactly one captain of the team.”

Truth of Quantified Statements

Table 7.1: Truth Conditions of Quantifiers

Quantifier	Truth Condition
$\forall x P(x)$	True if $P(x)$ is true for every element in the domain.
$\exists x P(x)$	True if $P(x)$ is true for at least one element in the domain.
$\exists! x P(x)$	True if exactly one element satisfies $P(x)$.

Negation of Quantifiers

The negation of quantified statements is given by

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x).$$

This means

”Not every student passed”

is equivalent to

”There exists at least one student who did not pass.”

Similarly,

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x).$$

This means

”There is no student absent”

is equivalent to

”Every student is present.”

7.8 Applications of Quantifiers

Quantifiers are widely used in mathematics, science, and engineering.

1. Mathematics

Used to define theorems, proofs, and mathematical properties.

2. Computer Programming

Used in loops, assertions, and formal program verification.

3. Artificial Intelligence

Knowledge representation and logical inference use quantified statements.

4. Database Systems

SQL queries use quantified conditions such as `EXISTS` and `FOR ALL`.

5. Digital Electronics

Formal verification of hardware specifications uses quantifiers.

6. Software Engineering

Used in software testing and model checking.

7. Cyber Security

Access-control policies are expressed using quantified rules.

8. Decision Support Systems

Business rules and constraints are modeled using quantified logic.

8 Predicate Calculus (Predicate Logic)

Definition

Predicate Calculus, also known as **Predicate Logic** or **First-Order Logic (FOL)**, is an extension of propositional logic that deals with predicates, variables, and quantifiers to represent and analyze statements involving objects and their properties.

Unlike propositional logic, which treats an entire statement as a single unit (True or False), predicate calculus describes the **internal structure** of statements by specifying objects, their properties, and relationships.

Mathematically, a predicate is represented as

$$P(x),$$

where

- P denotes a property or relation,
- x denotes an object (variable).

For example,

$$P(x) : \text{"}x \text{ is an even number."}$$

If

$$x = 4,$$

then

$$P(4) = \text{True.}$$

If

$$x = 7,$$

then

$$P(7) = \text{False.}$$

Components of Predicate Calculus

Predicate calculus consists of the following components.

1. Variables
2. Constants
3. Predicates
4. Functions
5. Quantifiers
6. Logical Connectives

1. Variables

Variables represent objects in the domain.

Examples:

$$x, y, z$$

2. Constants

Constants represent fixed objects.

Examples:

$$5, \pi, \text{India}, \text{Ravi}$$

3. Predicates

A predicate represents a property or relationship.

Example

$$P(x) :$$

" x is a prime number."

Example

$$L(x, y) :$$

" x likes y ."

4. Functions

Functions assign exactly one value to each object.

Example

$$f(x) = x + 2$$

5. Quantifiers

Predicate calculus uses two fundamental quantifiers.

$$\forall$$

Universal Quantifier (For every)

$$\exists$$

Existential Quantifier (There exists)

6. Logical Connectives

The logical connectives are

$$\neg, \wedge, \vee, \rightarrow, \leftrightarrow$$

8.1 Types of Predicate Calculus

Predicate calculus is generally classified into the following types.

1. First-Order Predicate Logic (First-Order Logic)
2. Higher-Order Predicate Logic

8.2 First-Order Predicate Logic

In First-Order Logic (FOL), quantifiers apply only to individual objects.

Example

$$\forall x \in \mathbb{R}, \quad x^2 \geq 0.$$

8.3 2. Higher-Order Predicate Logic

In Higher-Order Logic, quantifiers may apply to predicates, functions, or sets.

Example

$$\forall P, P(a).$$

Higher-order logic is mainly used in theorem proving and advanced mathematical logic.

Examples of Predicate Calculus

Example 1: Mathematics

Let

$$P(x) :$$

” x is an even number.”

Then

$$P(6) = \text{True}$$

$$P(9) = \text{False}$$

Example 2: University

Let

$$S(x) :$$

” x is a student.”

$$P(x) :$$

” x has passed the examination.”

Then

$$\forall x, S(x) \rightarrow P(x)$$

means

”Every student has passed the examination.”

Example 3: Banking

Let

$$C(x) :$$

” x is a customer.”

$$L(x) :$$

” x is eligible for a loan.”

Then

$$\exists x, C(x) \wedge L(x)$$

means

”There exists a customer who is eligible for a loan.”

Example 4: Healthcare

Let

$$P(x) :$$

” x is a patient.”

$$F(x) :$$

” x has a fever.”

Then

$$\exists x, P(x) \wedge F(x)$$

means

”There exists a patient who has a fever.”

Example 5: Smart Home

Let

$$M(x) :$$

” x is a motion sensor.”

$$A(x) :$$

” x activates the alarm.”

Then

$$\forall x, M(x) \rightarrow A(x)$$

means

”Every motion sensor activates the alarm.”

8.4 Applications of Predicate Calculus

Predicate calculus has numerous applications in science and engineering.

- 1. **Artificial Intelligence**
Knowledge representation, expert systems, and logical reasoning.
- 2. **Computer Science**
Programming language semantics and formal verification.
- 3. **Database Systems**
Query languages and relational databases.
- 4. **Software Engineering**
Specification and verification of software systems.
- 5. **Mathematics**
Writing definitions, axioms, and mathematical proofs.
- 6. **Digital Electronics**
Modeling logical circuits and hardware verification.
- 7. **Natural Language Processing**
Understanding and representing the meaning of human language.
- 8. **Robotics**
Decision making and intelligent planning.
- 9. **Cyber Security**
Modeling security policies and access-control rules.
- 10. **Expert Systems**
Medical diagnosis, financial decision making, and industrial automation.

8.5 Difference Between Propositional Logic and Predicate Calculus

Table 8.1: Propositional Logic vs. Predicate Calculus

Feature	Propositional Logic	Predicate Calculus
Basic Unit	Proposition	Predicate
Variables	Not used	Used
Quantifiers	Not available	Available
Expressiveness	Limited	More expressive
Example	$p \rightarrow q$	$\forall x (P(x) \rightarrow Q(x))$
Applications	Logic circuits	AI, databases, theorem proving

9 Inference Theory

Definition

Inference Theory is the branch of mathematical logic that studies the process of deriving valid conclusions from a given set of premises using established rules of inference.

In other words, inference theory provides systematic methods for determining whether a conclusion logically follows from one or more premises.

Suppose

$$P_1, P_2, \dots, P_n$$

are premises and

$$Q$$

is the conclusion. Then inference is represented as

$$P_1, P_2, \dots, P_n \vdash Q,$$

which is read as

“The conclusion Q is derived from the given premises.”

Inference theory forms the foundation of mathematical proofs, logical reasoning, artificial intelligence, and automated theorem proving.

Basic Components of Inference

An inference consists of the following components.

1. Premises
2. Rules of Inference
3. Conclusion

Premises

Premises are the statements assumed to be true.

Rules of Inference

Rules of inference are logical rules used to derive valid conclusions.

Conclusion

The conclusion is the statement obtained from the premises using valid reasoning.

9.1 Types of Inference

Inference is commonly classified into the following categories.

1. Deductive Inference
2. Inductive Inference
3. Abductive Inference

1. Deductive Inference

Definition

Deductive inference derives conclusions that are logically certain, provided the premises are true.

Example

Premise 1:

All engineering students must register for examinations.

Premise 2:

Ravi is an engineering student.

Conclusion:

Ravi must register for the examination.

Mathematically,

$$\forall x, E(x) \rightarrow R(x)$$

$$E(\text{Ravi})$$

Therefore,

$$R(\text{Ravi}).$$

2. Inductive Inference

Definition

Inductive inference derives a general conclusion from specific observations.

The conclusion is probable rather than logically certain.

Example

Observed facts:

- The first five students submitted the assignment on time.
- The next five students also submitted on time.

Conclusion:

Most students submit assignments on time.

3. Abductive Inference

Definition

Abductive inference selects the most plausible explanation for an observed fact.

Example

Observation:

The classroom floor is wet.

Possible conclusion:

It probably rained.

9.2 Rules of Inference in Propositional Logic

1. Modus Ponens

Rule

$$P \rightarrow Q$$

$$P$$

$$\therefore Q$$

Example

If a student pays the examination fee, then the hall ticket is issued.

The student paid the examination fee.

Therefore,

The hall ticket is issued.

2. Modus Tollens

Rule

$$P \rightarrow Q$$

$$\neg Q$$

$$\therefore \neg P$$

Example

If the ATM card is valid, then the transaction is approved.

The transaction was not approved.

Therefore,

The ATM card is not valid.

3. Hypothetical Syllogism

Rule

$$P \rightarrow Q$$

$$Q \rightarrow R$$

$$\therefore P \rightarrow R$$

Example

If a student studies,
then the student passes.

If the student passes,
then the student graduates.

Therefore,

If the student studies,
then the student graduates.

4. Disjunctive Syllogism

Rule

$$P \vee Q$$

$$\neg P$$

$$\therefore Q$$

Example

The meeting is either online or offline.

The meeting is not online.

Therefore,

The meeting is offline.

5. Addition Rule

Rule

$$P$$

$$\therefore P \vee Q$$

Example

The laboratory is open.

Therefore,

The laboratory is open or the library is open.

6. Simplification Rule

Rule

$$P \wedge Q$$

$$\therefore P$$

Example

A student has passed the examination and paid the tuition fee.

Therefore,

The student has passed the examination.

7. Conjunction Rule

Rule

$$P$$

$$Q$$

$$\therefore P \wedge Q$$

Example

The internet is available.

The server is operational.

Therefore,

The internet is available and the server is operational.

8. Resolution Rule**Rule**

$$P \vee Q$$

$$\neg P \vee R$$

$$\therefore Q \vee R$$

Example

Either the employee is working from home or attending a meeting.

Either the employee is not working from home or preparing a report.

Therefore,

The employee is attending a meeting or preparing a report.

9.3 Applications of Inference Theory**1. Artificial Intelligence**

Knowledge-based systems derive conclusions from stored facts using inference rules.

2. Expert Systems

Medical diagnosis systems infer diseases from observed symptoms.

3. Computer Programming

Compilers analyze logical conditions and optimize program execution.

4. Database Systems

Inference is used in query optimization and integrity checking.

5. Digital Electronics

Logic circuits operate according to inference principles.

6. Cyber Security

Authentication and access-control systems apply logical inference to verify user permissions.

7. Robotics

Autonomous robots make decisions based on sensor information and inference rules.

8. Legal Reasoning

Courts derive conclusions from legal facts and evidence.

9. Business Decision Making

Organizations infer trends and make strategic decisions from business data.

10. Mathematical Proofs

Inference rules provide the foundation for proving mathematical theorems.

Difference Between Deductive, Inductive, and Abductive Inference

Table 9.1: Types of Inference

Type	Nature	Example
Deductive	Certain conclusion	All humans are mortal. Ravi is human. Therefore, Ravi is mortal.
Inductive	Probable conclusion	Most observed birds can fly. Therefore, birds generally fly.
Abductive	Best explanation	The road is wet. Therefore, it probably rained.

10 Logic Gates and Their Operations

Definition

A **Logic Gate** is an electronic circuit that performs a logical operation on one or more binary inputs to produce a single binary output. Logic gates operate on binary values, namely

0 (False)

and

1 (True).

Logic gates are the fundamental building blocks of all digital electronic systems, including computers, calculators, mobile phones, communication devices, and embedded systems.

Basic Operations of Logic Gates

Each logic gate performs a specific logical operation based on Boolean algebra.

The basic operations are:

1. AND Operation
2. OR Operation
3. NOT Operation
4. NAND Operation
5. NOR Operation
6. XOR (Exclusive-OR) Operation
7. XNOR (Exclusive-NOR) Operation

10.1 Types of Logic Gates

Logic gates are classified into two categories.

1. Basic Logic Gates
2. Universal and Special Logic Gates

10.2 Basic Logic Gates

1. AND Gate
2. OR Gate
3. NOT Gate

(a) AND Gate

Definition

An AND gate produces an output of 1 only when **all** its inputs are 1.

$$Y = A \wedge B$$

Truth Table

A	B	$Y=A \wedge B$
0	0	0
0	1	0
1	0	0
1	1	1

Table 10.1: Truth Table of AND Gate

Example

A laboratory door opens only when

- the correct ID card is scanned, and
- the correct password is entered.

(b) OR Gate

Definition

An OR gate produces an output of 1 if at least one input is 1.

$$Y = A \vee B$$

Truth Table

A	B	$Y=A \vee B$
0	0	0
0	1	1
1	0	1
1	1	1

Table 10.2: Truth Table of OR Gate

Example

A fire alarm activates when

- smoke is detected, or
- excessive heat is detected.

(c) NOT Gate

Definition

A NOT gate produces the complement of the input.

$$Y = \neg A$$

Truth Table

A	Y
0	1
1	0

Table 10.3: Truth Table of NOT Gate

Example

A smart street light turns ON when daylight is absent.

10.3 Universal and Special Logic Gates

1. NAND Gate
2. NOR Gate
3. XOR Gate
4. XNOR Gate

(a) NAND Gate

Definition

A NAND gate is the complement of the AND gate.

$$Y = \overline{A \wedge B}$$

Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Table 10.4: Truth Table of NAND Gate

Example

A machine stops if all required safety conditions are satisfied.

(b) NOR Gate

Definition

A NOR gate is the complement of the OR gate.

$$Y = \overline{A \vee B}$$

Truth Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Table 10.5: Truth Table of NOR Gate

Example

A room remains silent only when both the television and music system are OFF.

(c) XOR Gate

Definition

An XOR (Exclusive-OR) gate produces an output of 1 only when the inputs are different.

$$Y = A \oplus B$$

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Table 10.6: Truth Table of XOR Gate

Example

A staircase light controlled by two switches changes state whenever either switch is toggled.

(d) XNOR Gate

Definition

An XNOR gate is the complement of the XOR gate.

It produces an output of 1 only when both inputs are the same.

$$Y = \overline{A \oplus B}$$

Truth Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Table 10.7: Truth Table of XNOR Gate

Example

A biometric authentication system grants access only when the stored fingerprint matches the scanned fingerprint.

Summary of Logic Gates

Table 10.8: Summary of Logic Gates

Gate	Boolean Expression	Operation
AND	$A \wedge B$	Both inputs must be 1
OR	$A \vee B$	At least one input is 1
NOT	$\neg A$	Complements the input
NAND	$\overline{A \wedge B}$	Complement of AND
NOR	$\overline{A \vee B}$	Complement of OR
XOR	$A \oplus B$	Inputs are different
XNOR	$\overline{A \oplus B}$	Inputs are identical

10.4 Applications of Logic Gates

Logic gates are widely used in modern digital systems.

1. **Computers** – Arithmetic Logic Units (ALUs), processors, and memory circuits.
2. **Digital Electronics** – Counters, multiplexers, encoders, decoders, and flip-flops.
3. **Communication Systems** – Error detection and data transmission circuits.
4. **Artificial Intelligence** – Hardware implementation of intelligent systems.
5. **Robotics** – Sensor-based control and decision-making circuits.
6. **Automotive Systems** – Airbags, anti-lock braking systems (ABS), and engine control units.
7. **Smart Homes** – Automatic lighting, security systems, and access control.
8. **Medical Equipment** – Patient monitoring systems and diagnostic devices.
9. **Industrial Automation** – PLCs, process control systems, and manufacturing automation.
10. **Consumer Electronics** – Mobile phones, televisions, calculators, washing machines, and microwave ovens.

11 Normal Forms

Definition

A **Normal Form** is a standardized representation of a logical expression using only logical connectives such as AND ($\wedge$), OR ($\vee$), and NOT ($\neg$). Every propositional logic expression can be converted into an equivalent normal form.

Normal forms simplify logical expressions and are widely used in digital electronics, computer science, artificial intelligence, and automated theorem proving.

Mathematically,

If

P

is a logical expression, then there exists an equivalent expression

Q

such that

$$P \equiv Q,$$

where Q is in a standard (normal) form.

Types of Normal Forms

There are two principal normal forms in propositional logic.

1. Conjunctive Normal Form (CNF)
2. Disjunctive Normal Form (DNF)

Some advanced texts also discuss the **Principal Conjunctive Normal Form (PCNF)** and the **Principal Disjunctive Normal Form (PDNF)**.

Conjunctive Normal Form (CNF)

Definition

A logical expression is said to be in **Conjunctive Normal Form (CNF)** if it is expressed as an AND ($\wedge$) of one or more clauses, where each clause is an OR ($\vee$) of literals.

General form:

$$(P_1 \vee P_2 \vee \dots) \wedge (Q_1 \vee Q_2 \vee \dots) \wedge \dots$$

where each literal is either a variable or its negation.

Example

$$(p \vee q) \wedge (\neg p \vee r)$$

is in CNF.

Example

Online Course Registration

A student can register only if

- the fee is paid or a scholarship is approved, and
- the prerequisite course is completed or permission is granted.

Let

p : Fee is paid

q : Scholarship is approved

r : Prerequisite completed

s : Permission granted

Then the condition is

$$(p \vee q) \wedge (r \vee s),$$

which is in CNF.

Problem 1

Convert the following logical expression into **Conjunctive Normal Form (CNF)**:

$$F = (A \Rightarrow B) \wedge (C \vee \bar{B})$$

Solution

Step 1: Write the given expression

$$F = (A \Rightarrow B) \wedge (C \vee \bar{B})$$

Step 2: Eliminate the implication

Recall the logical equivalence:

$$P \Rightarrow Q \equiv \bar{P} \vee Q.$$

Applying this rule,

$$A \Rightarrow B \equiv \bar{A} \vee B.$$

Substitute this into the given expression:

$$F = (\bar{A} \vee B) \wedge (C \vee \bar{B}).$$

Final Answer:

$$\boxed{F = (\bar{A} \vee B) \wedge (C \vee \bar{B})}$$

Therefore, the given logical expression is in **Conjunctive Normal Form (CNF)**.

Verification Using Truth Table

We verify that

$$(A \Rightarrow B) \wedge (C \vee \bar{B})$$

is logically equivalent to

$$(\bar{A} \vee B) \wedge (C \vee \bar{B}).$$

First, replace the implication:

$$A \Rightarrow B \equiv \bar{A} \vee B.$$

Construct the following truth table.

A	B	C	$\bar{A}$	$\bar{B}$	$A \Rightarrow B$	$C \vee \bar{B}$	$(A \Rightarrow B) \wedge (C \vee \bar{B})$	$(\bar{A} \vee B) \wedge (C \vee \bar{B})$
0	0	0	1	1	1	1	1	1
0	0	1	1	1	1	1	1	1
0	1	0	1	0	1	0	0	0
0	1	1	1	0	1	1	1	1
1	0	0	0	1	0	1	0	0
1	0	1	0	1	0	1	0	0
1	1	0	0	0	1	0	0	0
1	1	1	0	0	1	1	1	1

Since the values in the last two columns are identical for all possible combinations of A , B , and C , the two expressions are logically equivalent.

Hence,

$$(A \Rightarrow B) \wedge (C \vee \bar{B}) \equiv (\bar{A} \vee B) \wedge (C \vee \bar{B})$$

Therefore, the derived expression is the correct **Conjunctive Normal Form (CNF)** of the given logical expression.

Problem 2

Convert the following logical expression into **Conjunctive Normal Form (CNF)**:

$$F = (A \vee (B \Rightarrow C)) \wedge (\bar{C} \vee (A \Rightarrow B))$$

Solution

Step 1: Write the given expression

$$F = (A \vee (B \Rightarrow C)) \wedge (\bar{C} \vee (A \Rightarrow B))$$

Step 2: Eliminate the implications

Recall the logical equivalence:

$$P \Rightarrow Q \equiv \bar{P} \vee Q$$

Applying this rule,

$$B \Rightarrow C \equiv \bar{B} \vee C$$

and

$$A \Rightarrow B \equiv \bar{A} \vee B.$$

Substitute these into the given expression:

$$F = (A \vee (\bar{B} \vee C)) \wedge (\bar{C} \vee (\bar{A} \vee B))$$

Step 3: Apply the associative law of disjunction

Using the associative law,

$$A \vee (\bar{B} \vee C) \equiv A \vee \bar{B} \vee C,$$

and

$$\bar{C} \vee (\bar{A} \vee B) \equiv \bar{C} \vee \bar{A} \vee B.$$

Therefore,

$$F = (A \vee \bar{B} \vee C) \wedge (\bar{C} \vee \bar{A} \vee B).$$

Final Answer:

$$F = (A \vee \bar{B} \vee C) \wedge (\bar{C} \vee \bar{A} \vee B)$$

Therefore, the given logical expression is in **Conjunctive Normal Form (CNF)**.

Verification Using Truth Table

We verify that

$$(A \vee (B \Rightarrow C)) \wedge (\bar{C} \vee (A \Rightarrow B))$$

is logically equivalent to

$$(A \vee \bar{B} \vee C) \wedge (\bar{C} \vee \bar{A} \vee B).$$

First, replace the implications:

$$B \Rightarrow C \equiv \bar{B} \vee C, \quad A \Rightarrow B \equiv \bar{A} \vee B.$$

Construct the following truth table.

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$B \Rightarrow C$	$A \Rightarrow B$	$A \vee (B \Rightarrow C)$	$\bar{C} \vee (A \Rightarrow B)$	$A \vee \bar{B} \vee C$	$\bar{C} \vee \bar{A} \vee B$	$(A \vee (B \Rightarrow C)) \wedge (\bar{C} \vee (A \Rightarrow B))$	$(A \vee \bar{B} \vee C) \wedge (\bar{C} \vee \bar{A} \vee B)$
0	0	0	1	1	1	1	1	1	1	1	1	1	1
0	0	1	1	1	0	1	1	1	1	1	1	1	1
0	1	0	1	0	1	0	1	0	1	0	1	0	0
0	1	1	1	0	0	1	1	1	1	1	1	1	1
1	0	0	0	1	1	1	0	1	1	1	1	1	1
1	0	1	0	1	0	1	0	1	0	1	0	0	0
1	1	0	0	0	1	0	1	1	1	1	1	1	1
1	1	1	0	0	0	1	1	1	1	1	1	1	1

Since the values in the last two columns are identical for all possible combinations of A , B , and C , the two expressions are logically equivalent.

Hence,

$$(A \vee (B \Rightarrow C)) \wedge (\bar{C} \vee (A \Rightarrow B)) \equiv (A \vee \bar{B} \vee C) \wedge (\bar{C} \vee \bar{A} \vee B)$$

Therefore, the derived expression is Conjunctive Normal Form (CNF)

Problem 3: Security System

A security system should trigger an alarm if:

- The door is open ($D = 1$), OR
- The window is open ($W = 1$), AND
- The security guard is not present ($G = 0$).

Express the logical expression in **Conjunctive Normal Form (CNF)**.

Solution

Step 1: Define the logical variables

- D : Door is open.
- W : Window is open.
- G : Security guard is present.

Since the security guard is *not* present, we represent it by $\bar{G}$.

Step 2: Write the logical expression

The alarm is triggered if

- the door is open, OR
- the window is open AND the guard is not present.

Hence,

$$F = D \vee (W \wedge \bar{G}).$$

Step 3: Apply the distributive law

Recall the distributive law:

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R).$$

Applying this law,

$$F = (D \vee W) \wedge (D \vee \bar{G}).$$

Final Answer:

$$\boxed{F = (D \vee W) \wedge (D \vee \bar{G})}$$

Therefore, the CNF of the given security system is

$$\boxed{(D \vee W) \wedge (D \vee \bar{G})}.$$

Verification Using Truth Table

We verify that

$$D \vee (W \wedge \bar{G}) \equiv (D \vee W) \wedge (D \vee \bar{G})$$

by constructing the following truth table.

D	W	G	$\bar{G}$	$W \wedge \bar{G}$	$D \vee (W \wedge \bar{G})$	$D \vee W$	$D \vee \bar{G}$	$(D \vee W) \wedge (D \vee \bar{G})$
0	0	0	1	0	0	0	1	0
0	0	1	0	0	0	0	0	0
0	1	0	1	1	1	1	1	1
0	1	1	0	0	0	1	0	0
1	0	0	1	0	1	1	1	1
1	0	1	0	0	1	1	1	1
1	1	0	1	1	1	1	1	1
1	1	1	0	0	1	1	1	1

Since the values in the last two columns are identical for all possible combinations of D , W , and G , the two expressions are logically equivalent.

Hence,

$$D \vee (W \wedge \bar{G}) \equiv (D \vee W) \wedge (D \vee \bar{G})$$

Problem 4: Online Shopping Discount

An online store gives a discount if:

- The customer is a **student** ($S = 1$), OR
- The customer has a **coupon** ($C = 1$), BUT
- The discount is **not** given if the item is already on sale ($O = 1$).

Express the logical expression in **Conjunctive Normal Form (CNF)**.

Solution

Step 1: Define the logical variables

- S : Customer is a student.
- C : Customer has a coupon.
- O : Item is already on sale.

Since the discount is *not* given when the item is already on sale, we represent this condition by $\bar{O}$.

Step 2: Write the logical expression

The discount is given if

- the customer is a student, OR
- the customer has a coupon,

- and the item is not already on sale.

Hence,

$$F = (S \vee C) \wedge \bar{O}.$$

Step 3: Check whether the expression is in CNF

The expression

$$(S \vee C) \wedge \bar{O}$$

contains two clauses:

$$(S \vee C)$$

and

$$(\bar{O}).$$

The first clause is a disjunction of literals, while the second clause is a single literal (which is also a valid clause). Hence, the expression is already in **Conjunctive Normal Form (CNF)**.

Final Answer:

$$F = (S \vee C) \wedge \bar{O}$$

Therefore, the given logical expression is in **Conjunctive Normal Form (CNF)**.

Verification Using Truth Table

We verify that

$$(S \vee C) \wedge \bar{O}$$

is logically equivalent to its CNF expression

$$(S \vee C) \wedge \bar{O}.$$

Construct the following truth table.

S	C	O	$\bar{O}$	$S \vee C$	$(S \vee C) \wedge \bar{O}$	CNF
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	1	0	1	1	1	1
0	1	1	0	1	0	0
1	0	0	1	1	1	1
1	0	1	0	1	0	0
1	1	0	1	1	1	1
1	1	1	0	1	0	0

Since the values in the last two columns are identical for all possible combinations of S , C , and O , the original expression and the CNF expression are logically equivalent.

Hence,

$$(S \vee C) \wedge \bar{O} \equiv (S \vee C) \wedge \bar{O}$$

Therefore, the given expression is already in the correct **Conjunctive Normal Form (CNF)**.

11.1 Disjunctive Normal Form (DNF)

Definition

A logical expression is in **Disjunctive Normal Form (DNF)** if it is expressed as an OR ($\vee$) of one or more terms, where each term is an AND ($\wedge$) of literals.

General form:

$$(P_1 \wedge P_2) \vee (Q_1 \wedge Q_2) \vee \dots$$

Example 1

$$(p \wedge q) \vee (\neg p \wedge r)$$

is in DNF.

Example 2

Scholarship Eligibility

A student receives a scholarship if

- the student has high marks and low family income, or
- the student has a sports quota and excellent academic performance.

Let

p : High marks

q : Low family income

r : Sports quota

s : Excellent academic performance

Then

$$(p \wedge q) \vee (r \wedge s),$$

which is in DNF.

Problem 1

Convert the following logical expression into **Disjunctive Normal Form (DNF)**:

$$F = (p \vee q) \wedge r$$

Solution

Step 1: Write the given expression

$$F = (p \vee q) \wedge r$$

Step 2: Apply the distributive law

Recall the distributive law:

$$(P \vee Q) \wedge R \equiv (P \wedge R) \vee (Q \wedge R).$$

Applying this law,

$$F = (p \wedge r) \vee (q \wedge r).$$

Final Answer:

$$F = (p \wedge r) \vee (q \wedge r)$$

Therefore, the given logical expression is in **Disjunctive Normal Form (DNF)**.

Verification Using Truth Table

We verify that

$$(p \vee q) \wedge r$$

is logically equivalent to

$$(p \wedge r) \vee (q \wedge r).$$

Construct the following truth table.

p	q	r	$p \vee q$	$(p \vee q) \wedge r$	$p \wedge r$	$q \wedge r$	$(p \wedge r) \vee (q \wedge r)$
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	1	0	1	1
1	0	0	1	0	0	0	0
1	0	1	1	1	1	0	1
1	1	0	1	0	0	0	0
1	1	1	1	1	1	1	1

Since the values in the columns

$$(p \vee q) \wedge r$$

and

$$(p \wedge r) \vee (q \wedge r)$$

are identical for all possible combinations of p , q , and r , the two expressions are logically equivalent.
Hence,

$$(p \vee q) \wedge r \equiv (p \wedge r) \vee (q \wedge r)$$

Therefore, the derived expression is the correct **Disjunctive Normal Form (DNF)** of the given logical expression.

Problem 2

Convert the following Boolean expression into **Disjunctive Normal Form (DNF)**:

$$F = (A \wedge (B \vee C)) \vee (\neg B \wedge D)$$

Solution

Step 1: Write the given expression

$$F = (A \wedge (B \vee C)) \vee (\neg B \wedge D)$$

Step 2: Apply the distributive law

Recall the distributive law:

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R).$$

Applying this law,

$$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C).$$

Substitute this into the given expression:

$$F = ((A \wedge B) \vee (A \wedge C)) \vee (\neg B \wedge D).$$

Step 3: Apply the associative law of disjunction

Using the associative law,

$$((A \wedge B) \vee (A \wedge C)) \vee (\neg B \wedge D) \equiv (A \wedge B) \vee (A \wedge C) \vee (\neg B \wedge D).$$

Hence,

$$F = (A \wedge B) \vee (A \wedge C) \vee (\neg B \wedge D).$$

Final Answer:

$$F = (A \wedge B) \vee (A \wedge C) \vee (\neg B \wedge D)$$

Therefore, the given Boolean expression is in **Disjunctive Normal Form (DNF)**.

Verification Using Truth Table

We verify that

$$(A \wedge (B \vee C)) \vee (\neg B \wedge D)$$

is logically equivalent to

$$(A \wedge B) \vee (A \wedge C) \vee (\neg B \wedge D).$$

Construct the following truth table.

A	B	C	D	$B \vee C$	$A \wedge (B \vee C)$	$\neg B$	$\neg B \wedge D$	$A \wedge B$	$A \wedge C$	$(A \wedge (B \vee C)) \vee (\neg B \wedge D)$	$(A \wedge B) \vee (A \wedge C) \vee (\neg B \wedge D)$
0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	1	0	0	1	1	0	0	1	1
0	0	1	0	1	0	1	0	0	0	0	0
0	0	1	1	1	0	1	1	0	0	1	1
0	1	0	0	1	0	0	0	0	0	0	0
0	1	0	1	1	0	0	0	0	0	0	0
0	1	1	0	1	0	0	0	0	0	0	0
0	1	1	1	1	0	0	0	0	0	0	0
1	0	0	0	0	0	1	0	0	0	0	0
1	0	0	1	0	0	1	1	0	0	1	1
1	0	1	0	1	1	1	0	0	1	1	1
1	0	1	1	1	1	1	1	0	1	1	1
1	1	0	0	1	1	0	0	1	0	1	1
1	1	0	1	1	1	0	0	1	0	1	1
1	1	1	0	1	1	0	0	1	1	1	1
1	1	1	1	1	1	0	0	1	1	1	1

Since the values in the last two columns are identical for all possible combinations of A , B , C , and D , the two expressions are logically equivalent.

Hence,

$$(A \wedge (B \vee C)) \vee (\neg B \wedge D) \equiv (A \wedge B) \vee (A \wedge C) \vee (\neg B \wedge D)$$

Therefore, the derived expression is the correct **Disjunctive Normal Form (DNF)** of the given Boolean expression.

Problem 3: Smart Home Light Control

A smart home light turns ON if:

- It is evening ($E = 1$) AND someone is at home ($H = 1$), OR
- It is dark ($D = 1$) AND the motion sensor detects movement ($M = 1$).

Express the logical expression in **Disjunctive Normal Form (DNF)**.

Solution

Step 1: Define the logical variables

- E : It is evening.
- H : Someone is at home.
- D : It is dark.
- M : The motion sensor detects movement.

Step 2: Write the logical expression

The smart home light turns ON if

- it is evening and someone is at home, OR
- it is dark and the motion sensor detects movement.

Hence,

$$F = (E \wedge H) \vee (D \wedge M).$$

Final Answer:

$$F = (E \wedge H) \vee (D \wedge M)$$

Therefore, the given logical expression is already in **Disjunctive Normal Form (DNF)**.

Verification Using Truth Table

We verify that

$$(E \wedge H) \vee (D \wedge M)$$

is logically equivalent to its DNF expression

$$(E \wedge H) \vee (D \wedge M).$$

Construct the following truth table.

E	H	D	M	$E \wedge H$	$D \wedge M$	$(E \wedge H) \vee (D \wedge M)$	$(E \wedge H) \vee (D \wedge M)$
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	1	1	1
0	1	0	0	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	0	0	0	0	0
0	1	1	1	0	1	1	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0
1	0	1	0	0	0	0	0
1	0	1	1	0	1	1	1
1	1	0	0	1	0	1	1
1	1	0	1	1	0	1	1
1	1	1	0	1	0	1	1
1	1	1	1	1	1	1	1

Since the values in the last two columns are identical for all possible combinations of E , H , D , and M , the original expression and the DNF expression are logically equivalent.

Hence,

$$(E \wedge H) \vee (D \wedge M) \equiv (E \wedge H) \vee (D \wedge M)$$

Therefore, the given expression is already in the correct **Disjunctive Normal Form (DNF)**.

Problem 4: Online Exam Eligibility

A student is eligible to take an online exam if:

- The student has registered ($R = 1$) AND paid fees ($F = 1$), OR
- The student has been granted special permission ($P = 1$).

Express the logical expression in **Disjunctive Normal Form (DNF)**.

Solution

Step 1: Define the logical variables

- R : The student has registered.
- F : The student has paid the fees.
- P : The student has been granted special permission.

Step 2: Write the logical expression

The student is eligible if

- the student has registered and paid the fees, OR
- the student has special permission.

Hence,

$$F = (R \wedge F) \vee P.$$

Final Answer:

$$F = (R \wedge F) \vee P$$

Therefore, the given logical expression is already in **Disjunctive Normal Form (DNF)**.

Verification Using Truth Table

We verify that

$$(R \wedge F) \vee P$$

is logically equivalent to its DNF expression

$$(R \wedge F) \vee P.$$

Construct the following truth table.

R	F	P	$R \wedge F$	$(R \wedge F) \vee P$	DNF
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	0	0	0
0	1	1	0	1	1
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	1	1
1	1	1	1	1	1

Since the values in the last two columns are identical for all possible combinations of R , F , and P , the original expression and the DNF expression are logically equivalent.

Hence,

$$(R \wedge F) \vee P \equiv (R \wedge F) \vee P$$

Therefore, the given expression is already in the correct **Disjunctive Normal Form (DNF)**.

11.2 Principal Disjunctive Normal Form (PDNF)

Minterms

- **Definition:** A **minterm** is a product (AND) of all variables in the function, where each variable appears in either true (uncomplemented) or false (complemented) form.
- **Condition:** A minterm evaluates to 1 for exactly one combination of input variables and 0 for all others.
- **Representation:**

Case 1: If variables are A, B then minterms are

A	B	Minterm	Boolean Expression
0	0	m_0	$A'B'$
0	1	m_1	$A'B$
1	0	m_2	AB'
1	1	m_3	AB

Case 2: If variables are A, B, C then minterms are

A	B	C	Minterm	Boolean Expression
0	0	0	m_0	$A'B'C'$
0	0	1	m_1	$A'B'C$
0	1	0	m_2	$A'BC'$
0	1	1	m_3	$A'BC$
1	0	0	m_4	$AB'C'$
1	0	1	m_5	$AB'C$
1	1	0	m_6	ABC'
1	1	1	m_7	ABC

- **Used in:** Sum of Products (SOP) form, where the Boolean function is expressed as the sum (OR) of minterms.
- Example:

$$F(A, B) = A'B + AB' = m_1 + m_2$$

This means the function is written as a sum (OR) of two minterms.

Definition

The **Principal Disjunctive Normal Form (PDNF)**, also called the **Canonical Disjunctive Normal Form**, is a disjunction of **minterms** in which every variable appears exactly once, either complemented or uncomplemented.

Example

For

$$F(p, q) = p \rightarrow q,$$

its PDNF is

$$(\neg p \wedge \neg q) \vee (\neg p \wedge q) \vee (p \wedge q).$$

Example

A smart-home controller that turns on a device for every valid combination of sensors can be represented using PDNF.

Note

- CDNF uses all minterms corresponding to the function's truth table rows where the output = 1.
-

$$F = \sum m_i$$

where m_i are the minterms (product terms) for which the output is 1.

Problem 1:

Find the **Canonical Disjunctive Normal Form (CDNF)** of

$$f(A, B) = A + B$$

using logical laws and verify the result using a truth table.

Solution

Method 1: Using logical laws

Step 1: Write the given Boolean function

$$f(A, B) = A \vee B$$

Step 2: Introduce the missing variables

To obtain the Canonical Disjunctive Normal Form (CDNF), every product term (minterm) must contain all the variables exactly once.

Use the identity

$$X \equiv X \wedge T \equiv X \wedge (Y \vee \bar{Y}).$$

Apply this identity to both terms.

$$A \equiv A \wedge T \equiv A \wedge (B \vee \bar{B})$$

and

$$B \equiv B \wedge T \equiv B \wedge (A \vee \bar{A})$$

Hence,

$$\begin{aligned} f(A, B) &= A \vee B \\ &= (A \wedge (B \vee \bar{B})) \vee (B \wedge (A \vee \bar{A})). \end{aligned}$$

Apply the distributive law

$$f(A, B) = (A \wedge B) \vee (A \wedge \bar{B}) \vee (A \wedge B) \vee (\bar{A} \wedge B).$$

Remove the repeated term using the idempotent law

$$X \vee X = X.$$

Therefore,

$$f(A, B) = (A \wedge \bar{B}) \vee (\bar{A} \wedge B) \vee (A \wedge B).$$

Step 3: Write the CDNF

Each product term contains all the variables exactly once.

Hence, the Canonical Disjunctive Normal Form is

$$f(A, B) = (\bar{A} \wedge B) \vee (A \wedge \bar{B}) \vee (A \wedge B)$$

Method 2: Using Truth Table

Step 1: Write the truth table

A	B	$f(A, B) = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

Step 2: Identify the rows where the function value is 1

The function is equal to 1 for the following input combinations:

$$(0, 1), \quad (1, 0), \quad (1, 1).$$

Step 3: Write the corresponding minterms

For each row where the function value is 1:

(A, B)	Minterm	Decimal Index
(0, 1)	$\bar{A}B$	m_1
(1, 0)	$A\bar{B}$	m_2
(1, 1)	AB	m_3

Step 4: Form the CDNF

The Canonical Disjunctive Normal Form is obtained by taking the logical OR (sum) of all the minterms.

$$f(A, B) = \bar{A}B + A\bar{B} + AB$$

or, in summation notation,

$$f(A, B) = \Sigma m(1, 2, 3).$$

Problem 2

Find the **Canonical Disjunctive Normal Form (CDNF)** of

$$f(X, Y, Z) = X + YZ$$

using logical laws and verify the result using a truth table.

Solution

Method 1: Using logical laws

Step 1: Write the given Boolean function

$$f(X, Y, Z) = X \vee (Y \wedge Z)$$

Step 2: Introduce the missing variables

To obtain the Canonical Disjunctive Normal Form (CDNF), every product term (minterm) must contain all the variables exactly once.

Use the identity

$$X \equiv X \wedge T \equiv X \wedge (Y \vee \bar{Y}) \wedge (Z \vee \bar{Z}).$$

Also,

$$Y \wedge Z \equiv (Y \wedge Z) \wedge (X \vee \bar{X}).$$

Hence,

$$\begin{aligned} f(X, Y, Z) &= X \vee (Y \wedge Z) \\ &= \left(X \wedge (Y \vee \bar{Y}) \wedge (Z \vee \bar{Z}) \right) \vee \left((Y \wedge Z) \wedge (X \vee \bar{X}) \right). \end{aligned}$$

Step 3: Apply the distributive law

Applying the distributive law,

$$\begin{aligned} f(X, Y, Z) &= (X \wedge Y \wedge Z) \vee (X \wedge Y \wedge \bar{Z}) \vee (X \wedge \bar{Y} \wedge Z) \vee (X \wedge \bar{Y} \wedge \bar{Z}) \\ &\quad \vee (X \wedge Y \wedge Z) \vee (\bar{X} \wedge Y \wedge Z). \end{aligned}$$

Remove the repeated term using the idempotent law

$$X \vee X = X.$$

Therefore,

$$f(X, Y, Z) = (X \wedge \bar{Y} \wedge \bar{Z}) \vee (X \wedge \bar{Y} \wedge Z) \vee (X \wedge Y \wedge \bar{Z}) \vee (X \wedge Y \wedge Z) \vee (\bar{X} \wedge Y \wedge Z).$$

Step 4: Write the CDNF

Each conjunction contains all the variables exactly once.

Hence, the Canonical Disjunctive Normal Form is

$$f(X, Y, Z) = (X \wedge \bar{Y} \wedge \bar{Z}) \vee (X \wedge \bar{Y} \wedge Z) \vee (X \wedge Y \wedge \bar{Z}) \vee (X \wedge Y \wedge Z) \vee (\bar{X} \wedge Y \wedge Z)$$

Method 2: Using Truth Table

Step 1: Write the truth table

X	Y	Z	$f(X, Y, Z) = X + (YZ)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Step 2: Identify the rows where the function value is 1

The function is equal to 1 for the following input combinations:

$$(0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1).$$

Step 3: Write the corresponding minterms

For each row where the function value is 1:

(X, Y, Z)	Minterm	Decimal Index
$(0, 1, 1)$	$\bar{X}YZ$	m_3
$(1, 0, 0)$	$X\bar{Y}\bar{Z}$	m_4
$(1, 0, 1)$	$X\bar{Y}Z$	m_5
$(1, 1, 0)$	$XY\bar{Z}$	m_6
$(1, 1, 1)$	XYZ	m_7

Step 4: Form the CDNF

The Canonical Disjunctive Normal Form is obtained by taking the logical OR (sum) of all the minterms.

$$f(X, Y, Z) = (\bar{X}YZ) + (X\bar{Y}\bar{Z}) + (X\bar{Y}Z) + (XY\bar{Z}) + (XYZ)$$

or, in summation notation,

$$f(X, Y, Z) = \Sigma m(3, 4, 5, 6, 7).$$

Problem 3

A home security system is designed to activate an alarm whenever an **unauthorized person** attempts to enter the house through either the **door** or the **window**. If the person is authorized, the alarm remains OFF regardless of whether the door or window is opened.

Find the **Canonical Disjunctive Normal Form (CDNF)** of the alarm function using logical laws and verify the result using a truth table.

Let

- D : Door is opened.
- W : Window is opened.
- A : Person is authorized.
- F : Alarm is ON.

Solution

Method 1: Using logical laws

Step 1: Write the given Boolean function

Since the alarm sounds only when an unauthorized person enters through the door or the window,

$$F = (D \vee W) \wedge \bar{A}.$$

Step 2: Introduce the missing variables

To obtain the Canonical Disjunctive Normal Form (CDNF), every product term (minterm) must contain all the variables exactly once.

Expand each product term using the identity

$$X \equiv X \wedge (Y \vee \bar{Y}).$$

Hence,

$$\begin{aligned} F &= (D \vee W) \wedge \bar{A} \\ &= (D \wedge \bar{A}) \vee (W \wedge \bar{A}) \\ &= (D \wedge \bar{A} \wedge (W \vee \bar{W})) \vee (W \wedge \bar{A} \wedge (D \vee \bar{D})). \end{aligned}$$

Step 3: Apply the distributive law

Applying the distributive law,

$$F = (D \wedge W \wedge \bar{A}) \vee (D \wedge \bar{W} \wedge \bar{A}) \vee (D \wedge W \wedge \bar{A}) \vee (\bar{D} \wedge W \wedge \bar{A}).$$

Remove the repeated term using the idempotent law

$$X \vee X = X.$$

Therefore,

$$F = (\bar{D} \wedge W \wedge \bar{A}) \vee (D \wedge \bar{W} \wedge \bar{A}) \vee (D \wedge W \wedge \bar{A}).$$

Step 4: Write the CDNF

Each conjunction contains all the variables exactly once.

Hence, the Canonical Disjunctive Normal Form is

$$F = (\bar{D} \wedge W \wedge \bar{A}) \vee (D \wedge \bar{W} \wedge \bar{A}) \vee (D \wedge W \wedge \bar{A})$$

Method 2: Using Truth Table

Step 1: Write the truth table

D	W	A	$F(D, W, A) = (D + W)\bar{A}$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Step 2: Identify the rows where the function value is 1

The function is equal to 1 for the following input combinations:

$$(0, 1, 0), (1, 0, 0), (1, 1, 0).$$

Step 3: Write the corresponding minterms

For each row where the function value is 1:

(D, W, A)	Minterm	Decimal Index
$(0, 1, 0)$	$\bar{D}W\bar{A}$	m_2
$(1, 0, 0)$	$D\bar{W}\bar{A}$	m_4
$(1, 1, 0)$	$DW\bar{A}$	m_6

Step 4: Form the CDNF

The Canonical Disjunctive Normal Form is obtained by taking the logical OR (sum) of all the minterms.

$$F(D, W, A) = (\bar{D}W\bar{A}) + (D\bar{W}\bar{A}) + (DW\bar{A})$$

or, in summation notation,

$$F(D, W, A) = \Sigma m(2, 4, 6).$$

Problem 4: Student Scholarship Eligibility

A university awards a scholarship to a student if the student has **secured good academic performance or excelled in sports**, provided the student has **good attendance**. If the student does not have good attendance, the scholarship is not awarded.

Find the **Canonical Disjunctive Normal Form (CDNF)** of the scholarship eligibility function using logical laws and verify the result using a truth table.

Let

- A : Student has good academic performance.
- S : Student has excelled in sports.
- G : Student has good attendance.
- F : Student is eligible for the scholarship.

Solution

Method 1: Using logical laws

Step 1: Write the given Boolean function

The scholarship is awarded if the student has good academic performance or has excelled in sports, and also has good attendance.

$$F = (A \vee S) \wedge G.$$

Step 2: Introduce the missing variables

To obtain the Canonical Disjunctive Normal Form (CDNF), every product term (minterm) must contain all the variables exactly once.

Expand each product term using the identity

$$X \equiv X \wedge (Y \vee \bar{Y}).$$

Hence,

$$\begin{aligned}
F &= (A \vee S) \wedge G \\
&= (A \wedge G) \vee (S \wedge G) \\
&= (A \wedge G \wedge (S \vee \bar{S})) \vee (S \wedge G \wedge (A \vee \bar{A})).
\end{aligned}$$

Step 3: Apply the distributive law

Applying the distributive law,

$$F = (A \wedge S \wedge G) \vee (A \wedge \bar{S} \wedge G) \vee (A \wedge S \wedge G) \vee (\bar{A} \wedge S \wedge G).$$

Remove the repeated term using the idempotent law

$$X \vee X = X.$$

Therefore,

$$F = (\bar{A} \wedge S \wedge G) \vee (A \wedge \bar{S} \wedge G) \vee (A \wedge S \wedge G).$$

Step 4: Write the CDNF

Each conjunction contains all the variables exactly once.

Hence, the Canonical Disjunctive Normal Form is

$$F = (\bar{A} \wedge S \wedge G) \vee (A \wedge \bar{S} \wedge G) \vee (A \wedge S \wedge G)$$

or, in summation notation,

$$F(A, S, G) = \Sigma m(3, 5, 7).$$

Method 2: Using Truth Table

Step 1: Write the truth table

A	S	G	$F(A, S, G) = (A + S)G$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Step 2: Identify the rows where the function value is 1

The function is equal to 1 for the following input combinations:

$$(0, 1, 1), (1, 0, 1), (1, 1, 1).$$

Step 3: Write the corresponding minterms

For each row where the function value is 1:

(A, S, G)	Minterm	Decimal Index
$(0, 1, 1)$	$\bar{A}SG$	m_3
$(1, 0, 1)$	$A\bar{S}G$	m_5
$(1, 1, 1)$	ASG	m_7

Step 4: Form the CDNF

The Canonical Disjunctive Normal Form is obtained by taking the logical OR (sum) of all the minterms.

$$F(A, S, G) = (\bar{A}SG) + (A\bar{S}G) + (ASG)$$

or, in summation notation,

$$F(A, S, G) = \Sigma m(3, 5, 7).$$

11.3 Principal Conjunctive Normal Form (PCNF)

Maxterms

- **Definition:** A **maxterm** is a sum (OR) of all variables in the function, where each variable appears exactly once, either in true (uncomplemented) or false (complemented) form.
- **Condition:** A maxterm evaluates to 0 for exactly one combination of input variables and 1 for all other combinations.
- **Representation:**

Case 1: If variables are A, B , then the maxterms are

A	B	Maxterm	Boolean Expression
0	0	M_0	$A + B$
0	1	M_1	$A + B'$
1	0	M_2	$A' + B$
1	1	M_3	$A' + B'$

Case 2: If variables are A, B, C , then the maxterms are

A	B	C	Maxterm	Boolean Expression
0	0	0	M_0	$A + B + C$
0	0	1	M_1	$A + B + C'$
0	1	0	M_2	$A + B' + C$
0	1	1	M_3	$A + B' + C'$
1	0	0	M_4	$A' + B + C$
1	0	1	M_5	$A' + B + C'$
1	1	0	M_6	$A' + B' + C$
1	1	1	M_7	$A' + B' + C'$

- **Used in:** Product of Sums (POS) form, where the Boolean function is expressed as the product (AND) of maxterms.
- **Example:**

$$F(A, B) = (A + B') \wedge (A' + B) = M_1 \cdot M_2$$

This means the function is written as the product (AND) of two maxterms.

Definition

The **Principal Conjunctive Normal Form (PCNF)**, also called the **Canonical Conjunctive Normal Form (CCNF)**, is a conjunction (AND) of **maxterms** in which every variable appears exactly once, either complemented or uncomplemented.

Example

For

$$F(p, q) = p \rightarrow q,$$

its PCNF is

$$(p \vee q) \wedge (\neg p \vee q).$$

Application Example

An automated access-control system grants entry except for certain invalid combinations of inputs. Such systems are naturally represented using the Principal Conjunctive Normal Form (PCNF), where each invalid combination is represented by a maxterm.

Note

- CCNF uses all maxterms corresponding to the function's truth table rows where the output = 0.
-

$$F = \prod M_i$$

where M_i are the maxterms (sum terms) for which the output is 0.

- A Boolean function in CCNF is also called the **Canonical Product of Sums (Canonical POS)** form.

Problem 1

Find the **Canonical Conjunctive Normal Form (CCNF)** of

$$f(A, B) = A + B$$

using logical laws and verify the result using a truth table.

Solution

Method 1: Using logical laws

Step 1: Write the given Boolean function

$$f(A, B) = A \vee B$$

Step 2: Introduce the missing variables

To obtain the Canonical Conjunctive Normal Form (CCNF), every sum term (maxterm) must contain all the variables exactly once.

Use the identity

$$X \equiv X \vee F \equiv X \vee (Y \wedge \bar{Y}),$$

where $F = Y \wedge \bar{Y}$.

Apply this identity to the given function.

$$\begin{aligned} f(A, B) &= A \vee B \\ &= (A \vee B) \vee (B \wedge \bar{B}). \end{aligned}$$

Apply the distributive law

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R).$$

Hence,

$$f(A, B) = ((A \vee B) \vee B) \wedge ((A \vee B) \vee \bar{B}).$$

Using the idempotent law,

$$B \vee B = B,$$

we obtain

$$f(A, B) = (A \vee B) \wedge (A \vee B \vee \bar{B}).$$

Using the complement law,

$$B \vee \bar{B} = 1,$$

$$\begin{aligned} f(A, B) &= (A \vee B) \wedge (A \vee 1) \\ &= (A \vee B) \wedge 1 \\ &= A \vee B. \end{aligned}$$

Since the function contains only one maxterm,

$$\boxed{f(A, B) = A \vee B.}$$

Method 2: Using Truth Table

Step 1: Write the truth table

A	B	$f(A, B) = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

Step 2: Identify the rows where the function value is 0

The function is equal to 0 only for

$$(0, 0).$$

Step 3: Write the corresponding maxterm

For a maxterm,

- Write the variable uncomplemented if its value is 0.
- Write the variable complemented if its value is 1.

Therefore,

(A, B)	Maxterm	Decimal Index
$(0, 0)$	$A + B$	M_0

Step 4: Form the CCNF

The Canonical Conjunctive Normal Form is obtained by taking the logical AND (product) of all the maxterms. Since there is only one maxterm,

$$f(A, B) = A + B$$

or, in product notation,

$$f(A, B) = \Pi M(0).$$

Problem 2

Find the **Canonical Conjunctive Normal Form (CCNF)** of

$$f(A, B, C) = A\bar{B}$$

using logical laws and verify the result using a truth table.

Solution

Method 1: Using logical laws

Step 1: Write the given Boolean function

$$f(A, B, C) = A \wedge \bar{B}$$

Step 2: Introduce the missing variable

The variable C is missing. To obtain the Canonical Conjunctive Normal Form (CCNF), every maxterm must contain all the variables exactly once.

Use the identity

$$X \equiv X \vee (C \wedge \bar{C}).$$

Applying this identity,

$$\begin{aligned} f(A, B, C) &= A \wedge \bar{B} \\ &= (A \wedge \bar{B}) \vee (C \wedge \bar{C}). \end{aligned}$$

Apply the distributive law

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R).$$

Therefore,

$$f(A, B, C) = ((A \wedge \bar{B}) \vee C) \wedge ((A \wedge \bar{B}) \vee \bar{C}).$$

Again apply the distributive law

$$(X \wedge Y) \vee Z \equiv (X \vee Z) \wedge (Y \vee Z).$$

Then,

$$\begin{aligned} (A \wedge \bar{B}) \vee C &= (A \vee C) \wedge (\bar{B} \vee C), \\ (A \wedge \bar{B}) \vee \bar{C} &= (A \vee \bar{C}) \wedge (\bar{B} \vee \bar{C}). \end{aligned}$$

Hence,

$$f(A, B, C) = (A \vee C) \wedge (\bar{B} \vee C) \wedge (A \vee \bar{C}) \wedge (\bar{B} \vee \bar{C}).$$

Using the identity

$$(X \vee Y) \wedge (X \vee \bar{Y}) \equiv X,$$

we obtain

$$(A \vee C) \wedge (A \vee \bar{C}) = A,$$

and

$$(\bar{B} \vee C) \wedge (\bar{B} \vee \bar{C}) = \bar{B}.$$

Therefore,

$$f(A, B, C) = (A \vee C) \wedge (A \vee \bar{C}) \wedge (\bar{B} \vee C) \wedge (\bar{B} \vee \bar{C}) \\ = A \wedge \bar{B}.$$

The Canonical Conjunctive Normal Form is

$$f(A, B, C) = (A \vee B \vee C) \wedge (A \vee B \vee \bar{C}) \wedge (A \vee \bar{B} \vee C) \\ \wedge (A \vee \bar{B} \vee \bar{C}) \wedge (\bar{A} \vee \bar{B} \vee C) \wedge (\bar{A} \vee \bar{B} \vee \bar{C}).$$

Method 2: Using Truth Table

Step 1: Write the truth table

A	B	C	$f(A, B, C) = A\bar{B}$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

Step 2: Identify the rows where the function value is 0

The function is equal to 0 for

$$(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 1, 0), (1, 1, 1).$$

Step 3: Write the corresponding maxterms

For a maxterm,

- Write the variable uncomplemented if its value is 0.
- Write the variable complemented if its value is 1.

(A, B, C)	Maxterm	Decimal Index
$(0, 0, 0)$	$A + B + C$	M_0
$(0, 0, 1)$	$A + B + \bar{C}$	M_1
$(0, 1, 0)$	$A + \bar{B} + C$	M_2
$(0, 1, 1)$	$A + \bar{B} + \bar{C}$	M_3
$(1, 1, 0)$	$\bar{A} + \bar{B} + C$	M_6
$(1, 1, 1)$	$\bar{A} + \bar{B} + \bar{C}$	M_7

Step 4: Form the CCNF

The Canonical Conjunctive Normal Form is obtained by taking the logical AND (product) of all the maxterms.

$$f(A, B, C) = (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)(\bar{A} + \bar{B} + \bar{C}).$$

or, in product notation,

$$f(A, B, C) = \prod M(0, 1, 2, 3, 6, 7).$$

Problem 3: Home Security Alarm System

A home security system is designed to activate an alarm whenever an **unauthorized person** attempts to enter the house through either the **door** or the **window**. If the person is authorized, the alarm remains OFF regardless of whether the door or window is opened.

Find the **Canonical Conjunctive Normal Form (CCNF)** of the alarm function using logical laws and verify the result using a truth table.

Let

- D : Door is opened.
- W : Window is opened.
- A : Person is authorized.
- F : Alarm is ON.

Solution

Method 1: Using logical laws

Step 1: Write the given Boolean function

Since the alarm sounds only when an unauthorized person enters through the door or the window,

$$F = (D \vee W) \wedge \bar{A}.$$

Step 2: Identify the missing maxterms

For the Canonical Conjunctive Normal Form (CCNF), every sum term (maxterm) must contain all the variables exactly once. Expand each sum term using the identities

$$X \equiv X \vee (Y \wedge \bar{Y}),$$

and

$$\bar{A} \equiv \bar{A} \vee (D \wedge \bar{D}) \vee (W \wedge \bar{W}).$$

Hence,

$$\begin{aligned} F &= (D \vee W) \wedge \bar{A} \\ &= ((D \vee W) \vee (A \wedge \bar{A})) \wedge (\bar{A} \vee (D \wedge \bar{D}) \vee (W \wedge \bar{W})). \end{aligned}$$

Step 3: Apply the distributive law

Applying the distributive law,

$$F = (D \vee W \vee A) \wedge (D \vee W \vee \bar{A}) \\ \wedge (D \vee \bar{A} \vee W) \wedge (D \vee \bar{A} \vee \bar{W}) \\ \wedge (\bar{D} \vee \bar{A} \vee W) \wedge (\bar{D} \vee \bar{A} \vee \bar{W}).$$

Since

$$(D \vee W \vee \bar{A}) = (D \vee \bar{A} \vee W),$$

remove the repeated maxterm using the idempotent law

$$X \wedge X = X.$$

Therefore,

$$F = (D \vee W \vee A) \wedge (D \vee W \vee \bar{A}) \\ \wedge (D \vee \bar{A} \vee \bar{W}) \wedge (\bar{D} \vee \bar{A} \vee W) \\ \wedge (\bar{D} \vee \bar{A} \vee \bar{W}).$$

Step 4: Write the CCNF

Hence, the Canonical Conjunctive Normal Form is

$$F = (D \vee W \vee A) \wedge (D \vee W \vee \bar{A}) \\ \wedge (D \vee \bar{A} \vee \bar{W}) \wedge (\bar{D} \vee \bar{A} \vee W) \\ \wedge (\bar{D} \vee \bar{A} \vee \bar{W}).$$

or, in product notation,

$$F(D, W, A) = \prod M(0, 1, 3, 5, 7).$$

Method 2: Using Truth Table

Step 1: Write the truth table

D	W	A	$F(D, W, A) = (D + W)\bar{A}$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Step 2: Identify the rows where the function value is 0

The function is equal to 0 for

$$(0, 0, 0), (0, 0, 1), (0, 1, 1), (1, 0, 1), (1, 1, 1).$$

Step 3: Write the corresponding maxterms

For a maxterm,

- Write the variable uncomplemented if its value is 0.
- Write the variable complemented if its value is 1.

(D, W, A)	Maxterm	Decimal Index
$(0, 0, 0)$	$D + W + A$	M_0
$(0, 0, 1)$	$D + W + \bar{A}$	M_1
$(0, 1, 1)$	$D + \bar{W} + \bar{A}$	M_3
$(1, 0, 1)$	$\bar{D} + W + \bar{A}$	M_5
$(1, 1, 1)$	$\bar{D} + \bar{W} + \bar{A}$	M_7

Step 4: Form the CCNF

The Canonical Conjunctive Normal Form is obtained by taking the logical AND (product) of all the maxterms.

$$F(D, W, A) = (D + W + A)(D + W + \bar{A})(D + \bar{W} + \bar{A})(\bar{D} + W + \bar{A})(\bar{D} + \bar{W} + \bar{A}).$$

or, in product notation,

$$F(D, W, A) = \prod M(0, 1, 3, 5, 7).$$

Example 4: Fire Emergency Alert System

A fire emergency alert system activates the alarm whenever the **main power supply is available** or when both the **smoke detector** and the **heat detector** are activated simultaneously.

Find the **Canonical Conjunctive Normal Form (CCNF)** of the alarm function using logical laws and verify the result using a truth table.

Let

- P : Main power supply is available.
- S : Smoke detector is activated.
- H : Heat detector is activated.
- F : Alarm is activated.

Solution

Method 1: Using logical laws

Step 1: Write the given Boolean function

The alarm is activated if the main power supply is available or if both the smoke detector and heat detector are activated.

$$F = P \vee (S \wedge H).$$

Step 2: Convert into Product of Sums (POS)

Using the distributive law,

$$X \vee (Y \wedge Z) = (X \vee Y) \wedge (X \vee Z),$$

we obtain

$$F = (P \vee S) \wedge (P \vee H).$$

Step 3: Introduce the missing variables

To obtain the Canonical Conjunctive Normal Form (CCNF), every maxterm must contain all the variables exactly once. Use the identity

$$X \equiv X \vee (Y \wedge \bar{Y}).$$

Expand each sum term.

For the first sum term,

$$\begin{aligned} P \vee S &= (P \vee S) \vee (H \wedge \bar{H}) \\ &= (P \vee S \vee H) \wedge (P \vee S \vee \bar{H}). \end{aligned}$$

Similarly,

$$\begin{aligned} P \vee H &= (P \vee H) \vee (S \wedge \bar{S}) \\ &= (P \vee H \vee S) \wedge (P \vee H \vee \bar{S}). \end{aligned}$$

Since

$$P \vee H \vee S = P \vee S \vee H,$$

remove the repeated maxterm using

$$X \wedge X = X.$$

Step 4: Write the CCNF

Hence,

$$\begin{aligned} F &= (P \vee S \vee H) \wedge (P \vee S \vee \bar{H}) \\ &\quad \wedge (P \vee \bar{S} \vee H). \end{aligned}$$

or, in product notation,

$$F(P, S, H) = \prod M(0, 2, 4).$$

Method 2: Using Truth Table**Step 1: Write the truth table**

P	S	H	$F(P, S, H) = P + SH$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Step 2: Identify the rows where the function value is 0

The function is equal to 0 for

$$(0, 0, 0), (0, 0, 1), (0, 1, 0).$$

Step 3: Write the corresponding maxterms

For a maxterm,

- Write the variable uncomplemented if its value is 0.
- Write the variable complemented if its value is 1.

(P, S, H)	Maxterm	Decimal Index
$(0, 0, 0)$	$P + S + H$	M_0
$(0, 0, 1)$	$P + S + \bar{H}$	M_1
$(0, 1, 0)$	$P + \bar{S} + H$	M_2

Step 4: Form the CCNF

The Canonical Conjunctive Normal Form is obtained by taking the logical AND (product) of all the maxterms.

$$F(P, S, H) = (P + S + H)(P + S + \bar{H})(P + \bar{S} + H)$$

or, in product notation,

$$F(P, S, H) = \prod M(0, 1, 2).$$

11.4 Conversion of CNF to DNF and vice versa

Problem 1

Convert the following Boolean expression from **Conjunctive Normal Form (CNF)** to **Disjunctive Normal Form (DNF)**.

$$F = (A \vee B) \wedge (\bar{A} \vee C)$$

Solution

Step 1: Apply the distributive law

Using the distributive law,

$$(X \vee Y) \wedge (P \vee Q) = (X \wedge P) \vee (X \wedge Q) \vee (Y \wedge P) \vee (Y \wedge Q),$$

we obtain

$$F = (A \wedge \bar{A}) \vee (A \wedge C) \vee (B \wedge \bar{A}) \vee (B \wedge C).$$

Step 2: Simplify the expression

Using the complement law,

$$A \wedge \bar{A} = 0,$$

the expression becomes

$$F = 0 \vee (A \wedge C) \vee (B \wedge \bar{A}) \vee (B \wedge C).$$

Using the identity law,

$$0 \vee X = X,$$

we obtain

$$F = (A \wedge C) \vee (\bar{A} \wedge B) \vee (B \wedge C).$$

Hence, the equivalent **Disjunctive Normal Form (DNF)** is

$$F = (A \wedge C) \vee (\bar{A} \wedge B) \vee (B \wedge C).$$

Problem 2: Smart Irrigation System

A smart irrigation system automatically turns ON the water pump if the following conditions are satisfied:

- Either the soil is dry or rain is not detected.
- Either rain is not detected or the water tank is full.

1. Write the Boolean expression in CNF.
2. Convert the CNF into DNF.
3. Simplify the DNF using Boolean algebra.

Solution

Step 1: Construct the Boolean Expression

Let

D : Soil is dry,

R : Rain is not detected,

T : Water tank is full,

P : Water pump is ON.

From the given conditions,

$$P = (D \vee R) \wedge (R \vee T).$$

This expression is already in Conjunctive Normal Form (CNF).

Step 2: Convert CNF to DNF

Apply the Distributive Law,

$$(X \vee Y) \wedge (Z \vee W) = (X \wedge Z) \vee (X \wedge W) \vee (Y \wedge Z) \vee (Y \wedge W).$$

Therefore,

$$\begin{aligned} P &= (D \vee R) \wedge (R \vee T) \\ &= (D \wedge R) \vee (D \wedge T) \vee (R \wedge R) \vee (R \wedge T). \end{aligned}$$

Step 3: Simplify Using Boolean Laws

Using the Idempotent Law,

$$R \wedge R = R.$$

Hence,

$$P = (D \wedge R) \vee (D \wedge T) \vee R \vee (R \wedge T).$$

Apply the Absorption Law,

$$R \vee (D \wedge R) = R,$$

and

$$R \vee (R \wedge T) = R.$$

Therefore,

$$P = R \vee (D \wedge T).$$

Step 4: Final DNF

The required Disjunctive Normal Form is

$$P = R \vee (D \wedge T).$$

Problem 3

Convert the following Boolean expression from **Disjunctive Normal Form (DNF)** to **Conjunctive Normal Form (CNF)**.

$$F = (A \wedge B) \vee (\bar{A} \wedge C)$$

Solution

Step 1: Write the given DNF

$$F = (A \wedge B) \vee (\bar{A} \wedge C).$$

Step 2: Apply the distributive law

Using the distributive law,

$$(X \wedge Y) \vee Z = (X \vee Z) \wedge (Y \vee Z),$$

we obtain

$$\begin{aligned} F &= (A \wedge B) \vee (\bar{A} \wedge C) \\ &= (A \vee (\bar{A} \wedge C)) \wedge (B \vee (\bar{A} \wedge C)). \end{aligned}$$

Again applying the distributive law,

$$A \vee (\bar{A} \wedge C) = (A \vee \bar{A}) \wedge (A \vee C),$$

and

$$B \vee (\bar{A} \wedge C) = (B \vee \bar{A}) \wedge (B \vee C).$$

Hence,

$$\begin{aligned} F &= (A \vee \bar{A}) \wedge (A \vee C) \\ &\quad \wedge (B \vee \bar{A}) \wedge (B \vee C). \end{aligned}$$

Step 3: Simplify using Boolean laws

Using the Complement Law,

$$A \vee \bar{A} = 1.$$

Therefore,

$$F = 1 \wedge (A \vee C) \wedge (B \vee \bar{A}) \wedge (B \vee C).$$

Using the Identity Law,

$$1 \wedge X = X,$$

we obtain

$$F = (A \vee C) \wedge (\bar{A} \vee B) \wedge (B \vee C).$$

Step 4: Final CNF

Hence, the equivalent Conjunctive Normal Form is

$$F = (A \vee C) \wedge (\bar{A} \vee B) \wedge (B \vee C).$$

Problem 4: Automatic Emergency Notification System

An automatic emergency notification system sends an alert if either of the following conditions is satisfied:

- The fire detector is activated and the emergency switch is ON.
- The fire detector is activated and the smoke detector is activated.

1. Write the Boolean expression in DNF.
2. Convert the DNF into CNF.
3. Simplify the CNF using Boolean algebra.

Solution

Step 1: Construct the Boolean Expression

Let

F : Fire detector is activated,

E : Emergency switch is ON,

S : Smoke detector is activated,

A : Emergency alert is sent.

From the given conditions,

$$A = (F \wedge E) \vee (F \wedge S).$$

This expression is already in Disjunctive Normal Form (DNF).

Step 2: Convert DNF to CNF

Apply the Distributive Law,

$$(X \wedge Y) \vee (X \wedge Z) = X \wedge (Y \vee Z).$$

Therefore,

$$\begin{aligned} A &= (F \wedge E) \vee (F \wedge S) \\ &= F \wedge (E \vee S). \end{aligned}$$

To express the result in CNF, introduce the missing variable using the Identity Law,

$$F = F \vee (E \wedge \bar{E}) = (F \vee E) \wedge (F \vee \bar{E}).$$

Substituting,

$$A = ((F \vee E) \wedge (F \vee \bar{E})) \wedge (E \vee S).$$

Step 3: Simplify Using Boolean Laws

The expression is already in Product of Sums (CNF).

Hence,

$$A = (F \vee E) \wedge (F \vee \bar{E}) \wedge (E \vee S).$$

Step 4: Final CNF

The required Conjunctive Normal Form is

$$A = (F \vee E) \wedge (F \vee \bar{E}) \wedge (E \vee S).$$

11.5 Applications of Normal Forms

Normal forms such as Conjunctive Normal Form (CNF) and Disjunctive Normal Form (DNF) have numerous applications in mathematics, computer science, and engineering.

1. Digital Circuit Design

Boolean expressions are converted into CNF or DNF before implementing digital circuits using logic gates. These forms simplify the design of combinational circuits.

2. Artificial Intelligence

Knowledge-based systems and expert systems represent logical rules in normal forms to perform efficient logical reasoning and automated decision making.

3. Database Query Optimization

Database Management Systems (DBMS) transform complex SQL conditions into normal forms to improve query processing and retrieval efficiency.

4. Automated Theorem Proving

CNF is widely used in automated theorem provers and SAT solvers to verify mathematical theorems and logical statements.

5. Compiler Design

Compilers simplify and optimize logical expressions during code generation by converting them into equivalent normal forms.

6. Software and Hardware Verification

Formal verification tools use normal forms to check the correctness and reliability of software programs and digital hardware systems.

7. Decision Support and Expert Systems

Business rule engines, medical diagnosis systems, and intelligent decision-support systems represent logical conditions in normal forms to provide accurate and consistent decisions.

12 Minimization of Boolean function (Karnaugh Maps)

Definition

A **Karnaugh Map (K-Map)** is a graphical method used to simplify Boolean expressions by grouping adjacent cells containing 1's (for SOP/minterms) or 0's (for POS/maxterms). It provides a systematic technique for obtaining the minimum Boolean expression without extensive algebraic manipulation.

A Karnaugh Map was introduced by **Maurice Karnaugh** in 1953 and is widely used in digital logic design.

The simplified Boolean expression obtained using a Karnaugh Map requires fewer logic gates, thereby reducing the cost, complexity, and power consumption of digital circuits.

Minimization of Boolean function

- **Minimization** of a Boolean function means simplifying the function to its most compact form while keeping the same truth table (i.e., without changing the output).
- Simplification reduces the **number of gates, inputs, and hardware cost** in digital circuits.
- A **Karnaugh Map (K-Map)** is a graphical method to minimize Boolean expressions.
- It is a table-like representation of truth values, where adjacent cells differ in only **one variable** (Gray code ordering).
- By grouping **1's** (for SOP/minterms) or **0's** (for POS/maxterms), we eliminate redundant variables.

Principle of Karnaugh Maps

A Karnaugh Map represents the truth table of a Boolean function in a two-dimensional grid arranged in **Gray code order**. Adjacent cells differ in only one variable.

The simplification process consists of grouping adjacent 1's (or 0's) into rectangles containing

$$1, 2, 4, 8, 16, \dots$$

cells.

Only groups having sizes equal to powers of two are allowed.

12.1 Types of Karnaugh Maps

Karnaugh Maps are classified according to the number of variables.

1. Two-variable K-Map (it contains $2^2 = 4$ cells)
2. Three-variable K-Map (it contains $2^3 = 8$ cells)
3. Four-variable K-Map (it contains $2^4 = 16$ cells)
4. Five-variable K-Map (it contains $2^5 = 32$ cells)
5. Six-variable K-Map (it contains $2^6 = 64$ cells)

Among these, the 2-, 3-, and 4-variable Karnaugh Maps are the most commonly used in Engineering Mathematics and Digital Electronics.

12.2 Steps for Karnaugh Map (K-Map) Simplification

Step 1: Identify the Number of Variables

- Count the number of variables in the Boolean function.
- Select the appropriate K-map:
 - 2 variables $\rightarrow$ 4 cells
 - 3 variables $\rightarrow$ 8 cells
 - 4 variables $\rightarrow$ 16 cells

Step 2: Construct the K-map

- Draw the K-map corresponding to the number of variables.

- Label the rows and columns using **Gray Code** ordering so that adjacent cells differ by only one variable.

Step 3: Fill the K-map

- Enter 1's in the cells corresponding to the given minterms (for SOP minimization).
- Enter 0's in the remaining cells.
- If don't-care conditions are given, mark them as X . These may be included in groups if they help simplify the expression.

Step 4: Form Groups

- Group adjacent 1's (for SOP) or adjacent 0's (for POS).
- Each group must contain 1, 2, 4, 8, . . . cells (powers of 2).
- Form the largest possible groups.
- Use the fewest possible groups.
- Every 1 (or 0) must belong to at least one group.
- Overlapping groups are allowed if they lead to further simplification.
- Edge cells are adjacent (wrap-around property).
- Diagonal cells are **not** adjacent.

Step 5: Determine the Simplified Terms

- For each group,
 - retain the variables that remain constant throughout the group,
 - eliminate the variables that change within the group.
- Example:
 - If $A = 0$ and $B = 1$ remain constant while C changes, the simplified term is $\bar{A}B$.
 - If only $C = 1$ remains constant, the simplified term is C .

Step 6: Write the Simplified Boolean Expression

- Combine all simplified terms obtained from the groups.
- If grouping 1's, combine the terms using the OR (+) operator to obtain the minimal **Sum of Products (SOP)** expression.
- If grouping 0's, combine the clauses using the AND (·) operator to obtain the minimal **Product of Sums (POS)** expression.

From Truth Table to K-map (Idea)

A K-map rearranges truth table outputs so that adjacent cells differ in exactly one variable (Gray code). This makes it easy to group 1s in sizes of powers of 2 to simplify Boolean expressions.

- Variables label the K-map edges in Gray code order (e.g., 00, 01, 11, 10).
- Each cell corresponds to a minterm m_i (index i is the binary value of the variables).
- Adjacent means differing in *one* bit only; edges *wrap around* and corners can be adjacent.

2-Variable: Truth Table and K-map

Truth Table (A,B)

A	B	Minterm	Index
0	0	$A'B'$	m_0
0	1	$A'B$	m_1
1	0	AB'	m_2
1	1	AB	m_3

$A \backslash B$	0	1
0	m_0	m_1
1	m_2	m_3

- Truth table indices follow binary order: $AB = 00, 01, 10, 11 \Rightarrow m_0, \dots, m_3$.
- K-map places them so adjacent cells differ in only one variable: here any horizontal/vertical neighbor flips A or B .
- Wrap-around (left–right, top–bottom) also counts as adjacency.

3-Variable: Truth Table and K-map

A	B	C	Minterm	Index
0	0	0	$A'B'C'$	m_0
0	0	1	$A'B'C$	m_1
0	1	0	$A'BC'$	m_2
0	1	1	$A'BC$	m_3
1	0	0	$AB'C'$	m_4
1	0	1	$AB'C$	m_5
1	1	0	ABC'	m_6
1	1	1	ABC	m_7

$A \backslash BC$	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Explanation:

- Truth table index is binary ABC .
- K-map columns use Gray code on BC : 00, 01, 11, 10 so neighboring columns differ in one bit.
- Horizontal adjacency flips only B or C ; vertical adjacency flips only A ; edges wrap (first and last column are adjacent).

4-Variable: Truth Table and K-map

A	B	C	D	Minterm	Index
0	0	0	0	$A'B'C'D'$	m_0
0	0	0	1	$A'B'C'D$	m_1
0	0	1	0	$A'B'CD'$	m_2
0	0	1	1	$A'B'CD$	m_3
0	1	0	0	$A'BC'D'$	m_4
0	1	0	1	$A'BC'D$	m_5
0	1	1	0	$A'BCD'$	m_6
0	1	1	1	$A'BCD$	m_7
1	0	0	0	$AB'C'D'$	m_8
1	0	0	1	$AB'C'D$	m_9
1	0	1	0	$AB'CD'$	m_{10}
1	0	1	1	$AB'CD$	m_{11}
1	1	0	0	$ABC'D'$	m_{12}
1	1	0	1	$ABC'D$	m_{13}
1	1	1	0	$ABCD'$	m_{14}
1	1	1	1	$ABCD$	m_{15}

$AB \backslash CD$	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

Explanation:

- Truth table indices are binary $ABCD$ (m_0 to m_{15}).
- K-map uses Gray code for both AB (rows) and CD (columns): 00, 01, 11, 10.
- Each cell differs by one bit from its 4 neighbors (left, right, up, down), including wrap-around adjacency.
- Corner cells are pairwise adjacent (they differ in two *separate* wrap-around moves by one bit each).

12.3 Examples

Example 1:

Simplify the Boolean function $F(A, B, C) = \sum m(1, 2, 3, 5, 7)$.
using a 3-variable Karnaugh Map.

Solution

Step 1: Write the Given Minterms

The given minterms are

$$m(1, 2, 3, 5, 7).$$

Their corresponding binary values are

Minterm	A	B	C
m_1	0	0	1
m_2	0	1	0
m_3	0	1	1
m_5	1	0	1
m_7	1	1	1

Step 2: Draw the Karnaugh Map

Arrange the columns in Gray code order:

00, 01, 11, 10.

Fill the K-map with the given minterms.

$A \backslash BC$	00	01	11	10
0	0	1	1	1
1	0	1	1	0

Step 3: Form the Largest Possible Groups

Group 1:

Group the four adjacent cells

m_1, m_3, m_5, m_7 .

$A \backslash BC$	00	01	11	10
0	0	1	1	1
1	0	1	1	0

For this group,

- Variable $C = 1$ remains constant.
- Variables A and B change.

Therefore,

Group 1 = C .

Group 2:

Group the two adjacent cells

m_2, m_3 .

$A \backslash BC$	00	01	11	10
0	0	1	1	1
1	0	1	1	0

For this group,

- $A = 0 \Rightarrow \bar{A}$
- $B = 1$ remains constant.
- C changes.

Therefore,

$$\text{Group 2} = \bar{A}B.$$

Step 4: Write the Simplified Boolean Expression

Combine the simplified terms obtained from all the groups.

$$F = C + \bar{A}B.$$

Step 5: Verification

Verify the result using Boolean algebra.

$$\begin{aligned} F &= C + \bar{A}B \\ &= C(\bar{A} + A) + \bar{A}B \\ &= \bar{A}C + AC + \bar{A}B \\ &= \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + ABC + \bar{A}BC' \\ &= \sum m(1, 2, 3, 5, 7). \end{aligned}$$

Thus, the simplified expression is correct.

Final Answer

$$F(A, B, C) = C + \bar{A}B.$$

Example 2:

Simplify the Boolean function

$$F(A, B, C) = \sum m(0, 2, 4, 5, 6)$$

using a 3-variable Karnaugh Map.

Solution

Step 1: Write the Given Minterms

The given minterms are

$$m(0, 2, 4, 5, 6).$$

Their corresponding binary values are

Minterm	A	B	C
m_0	0	0	0
m_2	0	1	0
m_4	1	0	0
m_5	1	0	1
m_6	1	1	0

Step 2: Draw the Karnaugh Map

Arrange the columns in Gray code order:

$$00, 01, 11, 10.$$

Fill the K-map with the given minterms.

$A \backslash BC$	00	01	11	10
0	1	0	0	1
1	1	1	0	1

Step 3: Form the Largest Possible Groups

Group 1:

Group the four adjacent cells

$$m_0, m_2, m_4, m_6.$$

$A \backslash BC$	00	01	11	10
0	1	0	0	1
1	1	1	0	1

For this group,

- $C = 0 \Rightarrow \bar{C}$ remains constant.
- Variables A and B change.

Therefore,

$$\text{Group 1} = \bar{C}.$$

Group 2:

Group the two adjacent cells

$$m_4, m_5.$$

$A \backslash BC$	00	01	11	10
0	1	0	0	1
1	1	1	0	1

For this group,

- $A = 1$ remains constant.
- $B = 0 \Rightarrow \bar{B}$ remains constant.
- C changes.

Therefore,

$$\text{Group 2} = A\bar{B}.$$

Step 4: Write the Simplified Boolean Expression

Combine the simplified terms obtained from all the groups.

$$F = \bar{C} + A\bar{B}.$$

Step 5: Verification

Verify the result using Boolean algebra.

$$\begin{aligned}
F &= \bar{C} + A\bar{B} \\
&= \bar{C}(\bar{A} + A) + A\bar{B} \\
&= \bar{A}\bar{C} + A\bar{C} + A\bar{B} \\
&= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C} + A\bar{B}C \\
&= \sum m(0, 2, 4, 5, 6).
\end{aligned}$$

Thus, the simplified expression is verified.

Final Answer

$$F(A, B, C) = \bar{C} + A\bar{B}.$$

Example 3:

Minimize the Boolean function

$$F(A, B, C, D) = \sum m(0, 2, 5, 6, 7, 8, 10, 13, 15)$$

using a 4-variable Karnaugh Map.

Solution

Step 1: Write the Given Minterms

The given minterms are

$$m(0, 2, 5, 6, 7, 8, 10, 13, 15).$$

Their corresponding binary values are

Minterm	A	B	C	D
m_0	0	0	0	0
m_2	0	0	1	0
m_5	0	1	0	1
m_6	0	1	1	0
m_7	0	1	1	1
m_8	1	0	0	0
m_{10}	1	0	1	0
m_{13}	1	1	0	1
m_{15}	1	1	1	1

Step 2: Draw the 4-Variable Karnaugh Map

Arrange both rows and columns in Gray code order:

00, 01, 11, 10.

$AB \backslash CD$	00	01	11	10
00	1	0	0	1
01	0	1	1	1
11	0	1	1	0
10	1	0	0	1

Step 3: Form the Largest Possible Groups

Group 1:

Group the four corner cells

$$m_0, m_2, m_8, m_{10}.$$

$AB \backslash CD$	00	01	11	10
00	1	0	0	1
01	0	1	1	1
11	0	1	1	0
10	1	0	0	1

For this group,

- $B = 0 \Rightarrow \bar{B}$ remains constant.
- $D = 0 \Rightarrow \bar{D}$ remains constant.
- A and C change.

Therefore,

$$\text{Group 1} = \bar{B}\bar{D}.$$

Group 2:

Group the four cells

$$m_5, m_7, m_{13}, m_{15}.$$

$AB \backslash CD$	00	01	11	10
00	1	0	0	1
01	0	1	1	1
11	0	1	1	0
10	1	0	0	1

For this group,

- $B = 1$ remains constant.
- $D = 1$ remains constant.
- A and C change.

Therefore,

$$\text{Group 2} = BD.$$

Group 3:

Group the two adjacent cells

$$m_6, m_7.$$

$AB \backslash CD$	00	01	11	10
00	1	0	0	1
01	0	1	1	1
11	0	1	1	0
10	1	0	0	1

For this group,

- $A = 0 \Rightarrow \bar{A}$
- $B = 1$
- $C = 1$
- D changes.

Therefore,

$$\text{Group 3} = \bar{A}BC.$$

Step 4: Write the Simplified Boolean Expression

Combining all the group terms,

$$F = \bar{B}\bar{D} + BD + \bar{A}BC.$$

Step 5: Verification

The obtained expression covers all the given minterms

$$m(0, 2, 5, 6, 7, 8, 10, 13, 15)$$

and no additional minterms.

Hence, the simplified Boolean expression is correct.

Final Answer

$$F(A, B, C, D) = \bar{B}\bar{D} + BD + \bar{A}BC.$$

Example 4:

Minimize the Boolean function

$$F(A, B, C, D) = \sum m(1, 3, 4, 6, 9, 11, 12, 14)$$

using a 4-variable Karnaugh Map.

Solution

Step 1: Write the Given Minterms

The given minterms are

$$m(1, 3, 4, 6, 9, 11, 12, 14).$$

Their corresponding binary values are

Minterm	A	B	C	D
m_1	0	0	0	1
m_3	0	0	1	1
m_4	0	1	0	0
m_6	0	1	1	0
m_9	1	0	0	1
m_{11}	1	0	1	1
m_{12}	1	1	0	0
m_{14}	1	1	1	0

Step 2: Draw the 4-Variable Karnaugh Map

Arrange both rows and columns in Gray code order

00, 01, 11, 10.

$AB \backslash CD$	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	1	0	0	1
10	0	1	1	0

Step 3: Form the Largest Possible Groups

Group 1:

Group the four cells

m_1, m_3, m_9, m_{11} .

$AB \backslash CD$	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	1	0	0	1
10	0	1	1	0

For this group,

- $B = 0 \Rightarrow \bar{B}$ remains constant.
- $D = 1$ remains constant.
- A and C change.

Therefore,

$$\text{Group 1} = \bar{B}D.$$

Group 2:

Group the four cells

m_4, m_6, m_{12}, m_{14} .

$AB \backslash CD$	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	1	0	0	1
10	0	1	1	0

For this group,

- $B = 1$ remains constant.
- $D = 0 \Rightarrow \bar{D}$ remains constant.
- A and C change.

Therefore,

$$\text{Group 2} = B\bar{D}.$$

Step 4: Write the Simplified Boolean Expression

Combining the simplified terms obtained from all the groups,

$$F = \bar{B}D + B\bar{D}.$$

Step 5: Verification

The expression

$$F = \bar{B}D + B\bar{D}$$

covers exactly the minterms

$$m(1, 3, 4, 6, 9, 11, 12, 14)$$

and no other minterms.

Hence, the simplified Boolean expression is correct.

Final Answer

$$F(A, B, C, D) = \bar{B}D + B\bar{D}.$$

This may also be written as

$$F = B \oplus D,$$

where $\oplus$ denotes the Exclusive-OR (XOR) operation.

Example 5:

Use the Karnaugh map representation to find the minimal Product-of-Sums (POS) expression of the Boolean function

$$F(A, B, C) = \Pi M(0, 2, 5).$$

Solution

Step 1: Write the Given Maxterms

The given maxterms are

$$M(0, 2, 5).$$

These correspond to the following binary values.

Maxterm	A	B	C
M_0	0	0	0
M_2	0	1	0
M_5	1	0	1

Since the function is given in terms of maxterms, place **0**'s in these cells and **1**'s in the remaining cells.

Step 2: Draw the Karnaugh Map

Arrange the columns in Gray code order

$$00, 01, 11, 10.$$

$A \setminus BC$	00	01	11	10
0	0	1	1	0
1	1	0	1	1

Step 3: Form the Largest Possible Groups of 0's

Group 1:

Group the two adjacent cells

$$M_0, M_2.$$

$A \setminus BC$	00	01	11	10
0	0	1	1	0
1	1	0	1	1

(These two cells are adjacent because of the wrap-around property.)

For this group,

- $A = 0$ remains constant.
- $C = 0$ remains constant.
- B changes.

For POS simplification,

- Constant 0 $\rightarrow$ variable without complement.
- Constant 1 $\rightarrow$ complemented variable.

Therefore,

$\text{Group 1} = (A + C).$

Group 2:

The remaining zero is

M_5 .

$A \backslash BC$	00	01	11	10
0	0	1	1	0
1	1	0	1	1

Since it cannot be grouped with any other zero, it forms a single-cell group.

For

$$A = 1, \quad B = 0, \quad C = 1,$$

the corresponding maxterm is

$$(\bar{A} + B + \bar{C}).$$

Step 4: Write the Minimal POS Expression

Multiply all the obtained maxterms.

$$F = (A + C)(\bar{A} + B + \bar{C}).$$

Step 5: Verification

Expand the expression.

$$\begin{aligned} F &= (A + C)(\bar{A} + B + \bar{C}) \\ &= A\bar{A} + AB + A\bar{C} + \bar{A}C + BC + C\bar{C} \\ &= AB + A\bar{C} + \bar{A}C + BC. \end{aligned}$$

The function is equal to

$$\Sigma m(1, 3, 4, 6, 7),$$

which is equivalent to

$$\Pi M(0, 2, 5).$$

Hence, the minimal POS expression is verified.

Final Answer

$$F(A, B, C) = (A + C)(\bar{A} + B + \bar{C}).$$

Example 6:

Find the minimal Product-of-Sums (POS) expression for

$$F(A, B, C, D) = \Pi M(0, 2, 5, 7, 8, 10, 12, 14)$$

using a 4-variable Karnaugh Map.

Solution

Step 1: Write the Given Maxterms

The given maxterms are

$$M(0, 2, 5, 7, 8, 10, 12, 14).$$

Their corresponding binary values are

Maxterm	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
M_0	0	0	0	0
M_2	0	0	1	0
M_5	0	1	0	1
M_7	0	1	1	1
M_8	1	0	0	0
M_{10}	1	0	1	0
M_{12}	1	1	0	0
M_{14}	1	1	1	0

Since the function is given in terms of maxterms, place **0**'s in these cells and **1**'s in the remaining cells.

Step 2: Draw the Karnaugh Map

Arrange both rows and columns in Gray code order

$$00, 01, 11, 10.$$

$AB \backslash CD$	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	0	1	1	0
10	0	1	1	0

Step 3: Form the Largest Possible Groups of 0's

Group 1:

Group the four corner cells

$$M_0, M_2, M_8, M_{10}.$$

$AB \backslash CD$	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	0	1	1	0
10	0	1	1	0

For this group,

- $B = 0$ remains constant.
- $D = 0$ remains constant.
- A and C change.

Hence, the corresponding maxterm is

$$(B + D).$$

Group 2:

Group the four cells

$$M_8, M_{10}, M_{12}, M_{14}.$$

$AB \backslash CD$	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	0	1	1	0
10	0	1	1	0

Together with the corresponding cells in row $AB = 11$,

$$M_{12}, M_{14},$$

these form a group of four.

For this group,

- $A = 1$ remains constant.
- $D = 0$ remains constant.
- B and C change.

Hence, the corresponding maxterm is

$$(\bar{A} + D).$$

Group 3:

Group the two adjacent cells

$$M_5, M_7.$$

$AB \backslash CD$	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	0	1	1	0
10	0	1	1	0

For this group,

- $A = 0$ remains constant.
- $B = 1$ remains constant.
- $D = 1$ remains constant.
- C changes.

Hence, the corresponding maxterm is

$$(A + \bar{B} + \bar{D}).$$

Step 4: Write the Minimal POS Expression

Multiply the maxterms obtained from all the groups.

$$F = (B + D)(\bar{A} + D)(A + \bar{B} + \bar{D}).$$

Step 5: Verification

The obtained Product-of-Sums expression covers exactly the given maxterms

$$\Pi M(0, 2, 5, 7, 8, 10, 12, 14),$$

and therefore represents the required minimal POS expression.

Final Answer

$$F(A, B, C, D) = (B + D)(\bar{A} + D)(A + \bar{B} + \bar{D}).$$

Example: Fire Detection and Alarm System

A fire detection system activates an alarm if any one of the following conditions is satisfied:

- Smoke is detected while the temperature is high.
- Smoke is detected while a gas leak is detected.
- The temperature is high while a gas leak is detected.

Construct the Boolean function and simplify it using a 3-variable Karnaugh Map.

Solution

Let

S : Smoke is detected,

T : High temperature is detected,

G : Gas leakage is detected,

A : Fire alarm is activated.

Step 1: Construct the Boolean Function

From the given conditions,

$$A = ST + SG + TG.$$

Step 2: Determine the Minterms

The Boolean function is true for

$$(ST) + (SG) + (TG).$$

Hence,

$$A = \Sigma m(3, 5, 6, 7).$$

Step 3: Draw the Karnaugh Map

Arrange the columns in Gray code order

00, 01, 11, 10.

$S \backslash TG$	00	01	11	10
0	0	0	0	0
1	0	1	1	1

Step 4: Form the Largest Possible Groups

Group 1:

Group the cells

m_6, m_7 .

$S \backslash TG$	00	01	11	10
0	0	0	0	0
1	0	1	1	1

Here,

- $S = 1$
- $T = 1$
- G changes.

Therefore,

$$\text{Group 1} = ST.$$

Group 2:

Group the cells

m_5, m_7 .

$S \backslash TG$	00	01	11	10
0	0	0	0	0
1	0	1	1	1

Here,

- $S = 1$
- $G = 1$
- T changes.

Therefore,

$$\text{Group 2} = SG.$$

Group 3:

Group the cells

$$m_3, m_7.$$

$S \backslash TG$	00	01	11	10
0	0	0	0	0
1	0	1	<u>1</u>	1

Using the wrap-around property,

$$m_3(011) \text{ and } m_7(111)$$

are adjacent vertically.

Here,

- $T = 1$
- $G = 1$
- S changes.

Therefore,

$$\text{Group 3} = TG.$$

Step 5: Write the Simplified Boolean Expression

Combining all the groups,

$$A = ST + SG + TG.$$

Since no larger grouping is possible, this is the minimum expression.

Step 6: Verification

The simplified expression covers exactly the minterms

$$m(3, 5, 6, 7),$$

which correspond to all conditions where at least two fire indicators are active.

Final Answer

$$A = ST + SG + TG.$$

Example: Smart Irrigation System

A smart irrigation system automatically turns ON the water pump under the following conditions:

- The soil is dry and the water tank is full.
- The soil is dry and rain is not detected.
- The water tank is full and rain is not detected.

Construct the Boolean function and simplify it using a 3-variable Karnaugh Map.

Solution

Let

D : Soil is dry,

T : Water tank is full,

R : Rain is not detected,

P : Water pump is ON.

Step 1: Construct the Boolean Function

From the given conditions,

$$P = DT + DR + TR.$$

Step 2: Determine the Minterms

The Boolean function is true for

$$(D, T, R) = 011, 101, 110, 111.$$

Hence,

$$P = \Sigma m(3, 5, 6, 7).$$

Step 3: Draw the Karnaugh Map

Arrange the columns in Gray code order

$$00, 01, 11, 10.$$

$D \backslash TR$	00	01	11	10
0	0	0	1	0
1	0	1	1	1

Step 4: Form the Largest Possible Groups

Group 1:

Group the cells

$$m_6, m_7.$$

These cells have

$$D = 1, \quad T = 1,$$

while R changes.

Therefore,

$$\text{Group 1} = DT.$$

Group 2:

Group the cells

$$m_5, m_7.$$

These cells have

$$D = 1, \quad R = 1,$$

while T changes.

Therefore,

$$\text{Group 2} = DR.$$

Group 3:

Group the cells

$$m_3, m_7.$$

These cells have

$$T = 1, \quad R = 1,$$

while D changes.

Therefore,

$$\text{Group 3} = TR.$$

Step 5: Write the Simplified Boolean Expression

Combining all the group terms,

$$P = DT + DR + TR.$$

No larger grouping is possible; therefore, this is the minimum SOP expression.

Step 6: Verification

The simplified expression covers exactly the minterms

$$m(3, 5, 6, 7),$$

which correspond to the required operating conditions of the irrigation system.

Final Answer

$$P = DT + DR + TR.$$

12.4 Applications of Karnaugh Maps

Karnaugh Maps (K-Maps) are widely used in digital electronics and computer engineering to simplify Boolean functions before implementing digital circuits. Some important applications are given below.

Smart Home Lighting System

A smart home lighting system automatically turns ON the lights when motion is detected and the room is dark. The controller receives inputs from motion and light sensors. A Karnaugh Map is used to simplify the Boolean expression, thereby reducing the number of logic gates required for implementation.

Traffic Signal Control System

Modern traffic signal controllers use inputs such as vehicle sensors, pedestrian crossing buttons, and emergency vehicle detectors to determine signal changes. Karnaugh Maps simplify the control logic, resulting in faster and more efficient traffic management systems.

ATM Authentication System

An ATM grants access only when a valid ATM card, correct PIN, and successful biometric verification are provided. The authentication logic can be simplified using Karnaugh Maps before implementing the electronic control circuit.

Fire Detection and Alarm System

A fire alarm system activates when smoke, excessive heat, or gas leakage is detected. Karnaugh Maps minimize the Boolean expressions obtained from multiple sensors, reducing hardware complexity and improving system reliability.

Industrial Automation and PLC Systems

Programmable Logic Controllers (PLCs) are widely used in industries to control machines, conveyor belts, robotic arms, and manufacturing processes. Karnaugh Maps simplify the control logic, resulting in efficient automation with reduced hardware cost and power consumption.